

28

Chebyshev Approximation: Theory

28.1 THE DEFINITION OF CHEBYSHEV POLYNOMIALS

The Chebyshev¹ polynomials $T_n(x)$ can be defined in many ways. Perhaps the simplest is

$$T_n(x) = \cos(n \arccos x)$$

To see that these are polynomials, recall de Moivre's² theorem

$$\cos n\theta + i \sin n\theta = (\cos \theta + i \sin \theta)^n$$

Expanding the binomial and taking the real part, we get

$$\cos n\theta = \sum_{k=0}^{n/2} C(n, 2k) (-1)^k \cos^{n-2k}\theta \sin^{2k}\theta$$

But using

$$\sin^2 \theta = (1 - \cos^2 \theta)$$

we get

$$\cos n\theta = \text{polynomial of degree } n \text{ in } \cos \theta$$

¹ The $T_n(x)$ notation comes from the spelling Chebyshev (1821–1894) used for his name in French, Tchebychef.

² Abraham de Moivre (1667–1754). Of him Newton late in life said “Ask M. de Moivre, he knows these things better than I do.”

Now writing

$$\theta = \arccos x \quad \text{or} \quad x = \cos \theta$$

and using

$$\cos(\arccos x) \equiv x$$

we get the result.

Notice that

$$T_n(x) = T_{-n}(x) = T_{|n|}(x)$$

which is often useful in simplifying identities.

A second approach to the Chebyshev polynomials is via the ultraspherical polynomials defined in Sec. 26.2 with $\alpha = -\frac{1}{2}$. We have from there the orthogonality relations

$$\int_{-1}^1 \frac{T_m(x)T_n(x)}{\sqrt{1-x^2}} dx = 0 \quad m \neq n$$

and the corresponding differential equation

$$(1-x^2)T_n''(x) - xT_n'(x) + n^2T_n(x) = 0$$

A third approach to the Chebyshev polynomials is to use the transformation

$$\theta = \arccos x$$

and recognize that the Chebyshev polynomials are the trigonometric cosine functions in disguise (see Fig. 28.1.1). We can show directly that the set of functions $\{\cos m\theta\} (0 \leq \theta \leq \pi)$ are orthogonal. We have

$$\begin{aligned} \int_0^\pi \cos m\theta \cos n\theta d\theta &= \frac{1}{2} \int_0^\pi [\cos(m+n)\theta + \cos(m-n)\theta] d\theta \\ &= \frac{1}{2} \left[\frac{\sin(m+n)\theta}{m+n} + \frac{\sin(m-n)\theta}{m-n} \right]_0^\pi \quad m \neq n \\ &= \begin{cases} 0 & m \neq n \\ \pi/2 & m = n \neq 0 \\ \pi & m = n = 0 \end{cases} \end{aligned}$$

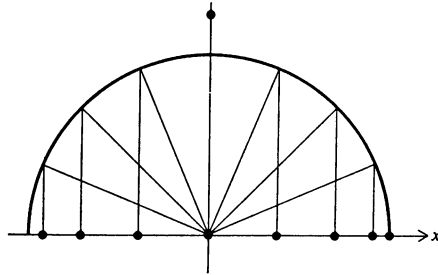


FIGURE 28.1.1
The Chebyshev sample points.

Transforming back from θ to x , we get

$$d\theta = \frac{-dx}{\sqrt{1-x^2}}$$

$$\int_{-1}^1 \frac{T_m(x)T_n(x)}{\sqrt{1-x^2}} dx = \begin{cases} 0 & m \neq n \\ \pi/2 & m = n \neq 0 \\ \pi & m = n = 0 \end{cases}$$

Thus we see that the first definition, the ultraspherical polynomials with $\alpha = -1/2$, and the Fourier cosine series all lead to the same set of polynomials.

PROBLEMS

28.1.1 Show that $T_m\{T_n(x)\} = T_{mn}(x)$.

28.1.2 If $g(x)$ has the inverse function $\arccos g(x)$, then the functions $f_n(x) = g(n \arccos g(x))$ satisfy $f_m\{f_n(x)\} = f_{mn}(x)$.

28.2 CHEBYSHEV POLYNOMIALS OVER A DISCRETE SET OF POINTS

We have regularly faced the problem of replacing a continuous interval by some set of points. It is an important fact, which we will prove in Chap. 31, that the Chebyshev polynomials $T_n(x)$ are orthogonal over the following discrete set of $N+1$ points x_i , equally spaced in θ ,

$$\theta_i = 0, \frac{\pi}{N}, \frac{2\pi}{N}, \dots, \frac{(N-1)\pi}{N}, \pi$$

where (see Fig. 28.1.1)

$$x_i = \cos \theta_i$$

We have

$$\frac{1}{2} T_m(-1)T_n(-1) + \sum_{i=2}^{N-1} T_m(x_i)T_n(x_i) + \frac{1}{2} T_m(1)T_n(1) = \begin{cases} 0 & m \neq n \\ N/2 & m = n \neq 0 \\ N & m = n = 0 \end{cases}$$

The $T_m(x)$ are also orthogonal over the following N points t_i , equally spaced in θ ,

$$\theta_i = \frac{\pi}{2N}, \frac{3\pi}{2N}, \frac{5\pi}{2N}, \dots, \frac{(2N-1)\pi}{2N}$$

and

$$\sum_{i=1}^N T_m(t_i) T_n(t_i) = \begin{cases} 0 & m \neq n \\ N/2 & m = n \neq 0 \\ N & m = n = 0 \end{cases} \quad t_i = \cos \theta_i$$

The set of points t_i are clearly the midpoints in θ of the first case.

The unequal spacing of the points in x_i (or t_i) compensates for the weight factor

$$w(x) = \frac{1}{\sqrt{1-x^2}}$$

in the continuous case.

28.3 FIRST PROPERTIES OF THE CHEBYSHEV POLYNOMIALS

The Chebyshev polynomials have a number of exceptional properties. First, they are orthogonal polynomials and therefore have all the properties of orthogonal polynomials. In particular, they have the three-term recurrence relation

$$T_{n+1}(x) - 2xT_n(x) + T_{n-1}(x) = 0$$

which is really the following trigonometric identity in disguise:

$$\cos(n+1)\theta + \cos(n-1)\theta = 2 \cos \theta \cos n\theta$$

It is important to note that the coefficients of this equation do not depend on n .

Using this relation, we construct the first few Chebyshev polynomials

$$T_0(x) = 1$$

$$T_1(x) = x$$

$$T_2(x) = 2x^2 - 1$$

$$T_3(x) = 4x^3 - 3x$$

$$T_4(x) = 8x^4 - 8x^2 + 1$$

$$T_5(x) = 16x^5 - 20x^3 + 5x$$

It is easy to see from the three-term recurrence relation that $T_k(x) = 2^{k-1}x^k + \dots$ and that the T_k are alternately odd and even polynomials.

Second, the zeros of $T_n(x)$ interlace those of $T_{n+1}(x)$ as is also obvious from the cosine representations.

Third, it is easy to go from a general polynomial to a Chebyshev expansion and back. Suppose we have the polynomial

$$\begin{aligned} & a_0 + a_1x + a_2x^2 + \cdots + a_Nx^N \\ &= a_0 + x\{a_1 + \cdots [a_{N-2} + x(a_{N-1} + a_Nx)] \cdots\} \end{aligned}$$

The inner parenthesis is

$$a_{N-1} + a_Nx \equiv a_{N-1}T_0 + a_NT_1$$

Multiplying by x , we get

$$\begin{aligned} x(a_{N-1} + a_Nx) &= a_{N-1}T_1 + a_N[\tfrac{1}{2}(T_0 + T_2)] \\ &= \frac{a_N}{2}T_0 + a_{N-1}T_1 + \frac{a_N}{2}T_2 \end{aligned}$$

Suppose in general that we have at some stage of converting from powers of x to Chebyshev polynomials

$$\{\cdots\cdots\} = b_0T_0 + b_1T_1 + b_2T_2 + \cdots + b_kT_k$$

Multiply by x and use

$$xT_k = \tfrac{1}{2}T_{k-1} + \tfrac{1}{2}T_{k+1}$$

Pictorially (see Fig. 28.3.1), if the top line has the coefficients at one stage, then we have the lower line at the next stage. Notice that after N stages, a_Nx^N goes to

$$\frac{a_N}{2^{N-1}}T_N(x) + \text{lower terms}$$

Studying this process carefully shows the basis for the claim that generally an expansion in Chebyshev polynomials converges much more rapidly than the

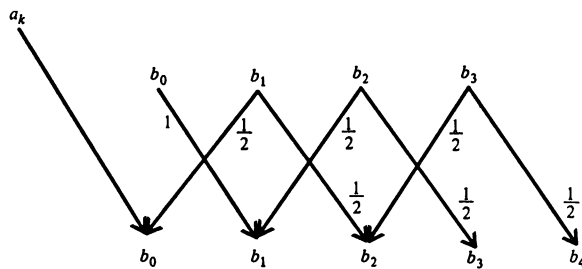


FIGURE 28.3.1

corresponding power series. It also shows that the sums of the coefficients in the two representations are the same!

To get from a Chebyshev expansion back to a polynomial, we can reverse the steps starting on the right.

PROBLEMS

28.3.1 Compute $T_6(x)$.

28.3.2 Prove that

$$T_{2k+1}(-x) = -T_{2k+1}(x)$$

and

$$T_{2k}(-x) = T_{2k}(x)$$

28.3.3 Convert

$$1 + x + x^2 + x^3 + x^4$$

to Chebyshev polynomials.

$$\text{Ans. } \frac{T_4}{8} + \frac{T_3}{4} + T_2 + \frac{7T_1}{4} + \frac{15}{8} T_0$$

28.3.4 Draw a flow diagram for converting from a polynomial to Chebyshev polynomials.

28.3.5 Draw a flow diagram for the reverse of Prob. 28.3.4.

28.3.6 Show that if the three-term relation has coefficients independent of n , then they are essentially the Chebyshev polynomials.

28.4 FURTHER PROPERTIES OF THE CHEBYSHEV POLYNOMIALS

The fact that the Chebyshev polynomials are the cosines in the disguise of

$$x = \cos \theta$$

means that they have another set of remarkable properties.

First, the polynomials are equal-ripple in the sense that the alternate maxima and minima are of the same size—exactly 1 in fact (since the maximum of $\cos \theta$ is 1). See Fig. 28.4.1.

Second, the orthogonality both over the continuous range and over the sets x_i and t_i of discrete points comes from the corresponding orthogonality of the cosines, which we will prove in Chap. 31. The importance of this should be appreciated because it means that in a very natural way we can approximate the

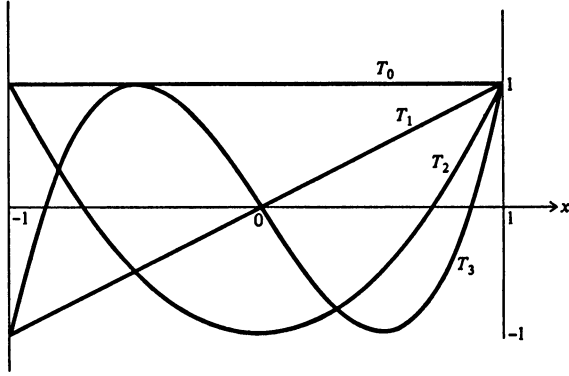


FIGURE 28.4.1

continuous case by simply doing the discrete case. This property is not enjoyed by the other classical sets of orthogonal polynomials; one can make a set of polynomials orthogonal over a set of points in the interval $-1 \leq x \leq 1$, but they will not have exactly the same analytic form as the Legendre polynomials.

Third, at the sample points which are the zeros of the N th Chebyshev polynomial, there is an *aliasing* effect (see Fig. 28.4.2) in that *at the sample points*, the polynomials of higher order look like lower-order Chebyshev polynomials. We can show this directly from the Chebyshev identity that follows from the trigonometric identity

$$\cos(m + N)\theta + \cos(m - N)\theta = 2 \cos m\theta \cos N\theta$$

or

$$T_{m+N}(x) + T_{m-N}(x) = 2T_m(x)T_N(x)$$

Holding N as fixed, at the zeros of $T_N(x) = 0$, that is, at

$$\cos N\theta_k = 0 \quad \theta_k = \frac{1}{N} \left[\frac{-\pi}{2} + k\pi \right] \quad k = 1, 2, \dots, N$$

we have

$$T_{m+N}(x_k) = -T_{N-m}(x_k)$$

For $m = N$ we have

$$T_{2N}(x_k) = -T_0(x_k) = -1$$

and in general it will follow that

$$T_{3N}(x_k) = 0 \quad T_{4N}(x_k) = 1 \quad T_{5N}(x_k) = 0 \quad T_{6N}(x_k) = -1 \quad \text{etc.}$$

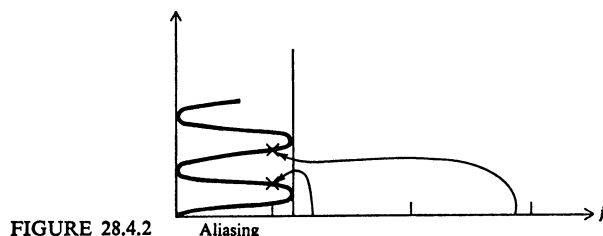


FIGURE 28.4.2

It should be evident that any order Chebyshev polynomial can be reduced, at the sample points x_k , which are the zeros of $T_N(x)$, to an equivalent Chebyshev polynomial with order j ($0 \leq j \leq N$). Figure 28.4.2 shows the reduction in a graphic form. Imagine the line of the index numbers of the order of the polynomial being folded like a ribbon, as shown in the figure. Then each index can be reduced to the smallest index, as shown in the figure.

The identity

$$T_m(x)T_n(x) = \frac{1}{2}[T_{m+n}(x) + T_{m-n}(x)]$$

has a second valuable use. It means that, similar to logs, products can be replaced by sums. Thus it is easy to multiply two series in Chebyshev polynomials—the products that arise are easily expressed as Chebyshev polynomials.

PROBLEMS

28.4.1 Draw a flow diagram for multiplying two Chebyshev expansions to get a third expansion.

28.5 THE CHEBYSHEV CRITERION

Chebyshev showed that, of all polynomials $p_n(x)$ of degree n having a leading coefficient equal to 1, the polynomial

$$\frac{T_n(x)}{2^{n-1}}$$

has the smallest least upper bound for its absolute value in the interval $-1 \leq x \leq 1$. Since the upper bound of $|T_n(x)|$ is 1, then this upper bound is $1/2^{n-1}$.

The proof of this remarkable property follows from an examination of the difference (see Fig. 28.5.1)

$$\phi_{n-1}(x) = \frac{T_n(x)}{2^{n-1}} - p_n(x)$$

which is a polynomial of degree $n - 1$, since x^n drops out. From the fact that $T_n(x)$ is $\cos n\theta$ in disguise, we see that in the interval $T_n(x)$ takes its extreme value $n + 1$ times, alternately positive and negative. If the extreme value of $p_n(x)$ is less than that of $T_n(x)/2^{n-1}$, then at these $n + 1$ extreme points $\phi_{n-1}(x)$ is alternately positive and negative; hence it must have n real zeros between these extremes. Since $\phi_{n-1}(x)$ is of degree $n - 1$, we conclude that $\phi_n(x) = 0$ and

$$p_n = \frac{T_n(x)}{2^{n-1}}$$

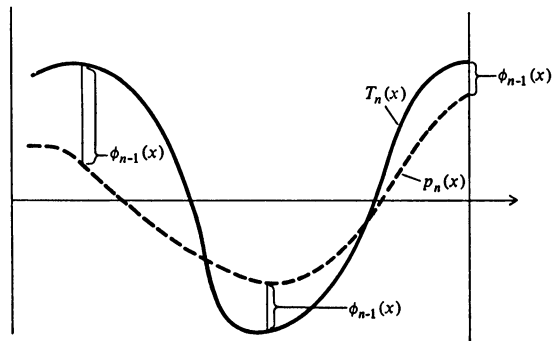


FIGURE 28.5.1

This property is of great interest in numerical computation. If some error can be expressed as a Chebyshev polynomial of degree n , then any other error which is a polynomial of degree n and which has the same leading coefficient will have, in the range $-1 \leq x \leq 1$, a greater extreme error than the Chebyshev error representation. Thus the expression *Chebyshev approximation* is associated with those approximations¹ which try to keep the maximum error to a minimum. This is sometimes called the *minimax principle*. Least-squares approximation keeps the *average* square of the error down, but in doing so isolated extreme errors are permitted; Chebyshev keeps the extreme errors down, but allows a larger average square error.

As a simple illustration, consider the problem of interpolating in the interval $-1 \leq x \leq 1$ with an exact-matching polynomial of degree n . The error term (Sec. 14.6) has the form

$$\frac{(x - x_1)(x - x_2) \cdots (x - x_{n+1})y^{(n+1)}(\bar{x})}{(n+1)!}$$

¹ Chebyshev approximation should not be confused with Chebyshev integration; he had many good ideas, some of which have been named after him.

If we want to minimize the maximum error due to the factor

$$(x - x_1)(x - x_2) \cdots (x - x_{n+1})$$

we can choose as sample points the zeros of $T_{n+1}(x) = 0$. Thus we have for the part of the error that we can control easily the equal-ripple polynomial which has the least maximum error.

In a sense the Chebyshev polynomials show how the fundamental theorem of algebra collapses as the degree of the polynomial increases. One interpretation of the fundamental theorem of algebra is that the powers of x are linearly independent over any interval, and to be definite we pick the interval $-1 \leq x \leq 1$. By picking the n th Chebyshev polynomial divided by 2^{n-1} we get a polynomial of degree n with leading coefficient 1, which is a reasonable way to normalize the polynomials when considering linear independence. From the fact that $|T_n(x)| \leq 1$ in the interval, it follows that

$$\frac{|T_n(x)|}{2^{n-1}} \leq \frac{1}{2^{n-1}} \quad -1 \leq x \leq 1$$

and for, say, $n = 21$, we have linear relation among the first 22 powers x^k ($k = 0, 1, 2, \dots, 21$) which is less than 10^{-6} in the whole interval. If the noise level of roundoff is considered to be 10^{-9} , then we can go to $n = 31$. In general for degree n we have a noise level below which a linear dependence among the powers of x occurs.

28.6 FURTHER IDENTITIES

We have already produced a number of identities among the Chebyshev polynomials. Because they are both orthogonal polynomials and the trigonometric functions $\cos nx$ in disguise, they satisfy an amazing number of relations, many of which are useful. The three-term relation shows how to multiply a Chebyshev polynomial by x (see Prob. 28.6.3 for an extension), and we have shown how to convert products of Chebyshev polynomials to sums of Chebyshev polynomials so that two Chebyshev series can be easily multiplied to get another Chebyshev series.

It is natural to ask how to differentiate and integrate. We begin with

$$\begin{aligned} T_{n+1}(x) &= \cos[(n+1)\arccos x] \\ \frac{1}{n+1} \frac{d[T_{n+1}(x)]}{dx} &= \frac{-\sin[(n+1)\arccos x]}{-\sqrt{1-x^2}} \end{aligned}$$

Write the corresponding equation for $n-1$ and subtract to get

$$\frac{1}{n+1} \frac{d[T_{n+1}(x)]}{dx} - \frac{1}{n-1} \frac{d[T_{n-1}(x)]}{dx} = \frac{\sin(n+1)\theta - \sin(n-1)\theta}{\sin \theta}$$

or

$$\frac{T'_{n+1}(x)}{n+1} - \frac{T'_{n-1}(x)}{n-1} = \frac{2 \cos n\theta \sin \theta}{\sin \theta} = 2T_n(x) \quad n \geq 2$$

$$T'_2(x) = 4T_1$$

$$T'_1(x) = T_0$$

$$T'_0(x) = 0$$

Thus we have the formulas for the differentiation of Chebyshev polynomials.

We can use these formulas to find the formulas for integrating Chebyshev polynomials:

$$\int T_n(x) dx = \frac{1}{2} \left[\frac{T_{n+1}(x)}{n+1} - \frac{T_{n-1}(x)}{n-1} \right] + C \quad n \geq 2$$

$$\int T_1(x) dx = \frac{T_2(x)}{4} + C$$

$$\int T_0(x) dx = T_1(x) + C$$

PROBLEMS

28.6.1 Show that

$$\frac{T'_{2n+1}}{2n+1} = 2(T_{2n} + T_{2n-2} + \cdots + T_2) + T_0$$

$$\frac{T'_{2n}}{2n} = 2(T_{2n-1} + T_{2n-3} + \cdots + T_1)$$

28.6.2 Show from Prob. 28.6.1 that

$$\frac{T'_k(x)}{k} = \sum_{j=0}^{n-1} T_{n-2k-1}$$

28.6.3 Show that

$$x^k T_n(x) = \frac{T_{n+k} + C(k, 1)T_{n+k-2} + C(k, 2)T_{n+k-4} + \cdots + T_{n-k}}{2^k}$$

28.6.4 Prove that

$$\begin{aligned} \frac{T_{2k+1}(x)}{x} &= 2T_{2k}(x) - 2T_{2k-2}(x) + \cdots \pm T_0 \\ &= (-1)^k \sum_{j=-k}^k (-1)^j T_{2j}(x) \end{aligned}$$

28.6.5 Integrate

$$\int x T_n(x) dx = \frac{T_{m+2}}{(m+1)(m+2)} - \frac{2m^2 T_m}{m(m^2-1)} + \frac{T_{m-2}}{(m-1)(m-2)} + C$$

28.7 THE SHIFTED CHEBYSHEV POLYNOMIALS

It is often convenient to use the interval $0 \leq x \leq 1$ instead of the interval $-1 \leq x \leq 1$. For this purpose the *shifted Chebyshev polynomials*

$$T_n^*(x) = T_n(2x - 1)$$

are used. Thus we have

$$\begin{aligned} T_0^*(x) &= 1 \\ T_1^*(x) &= 2x - 1 \\ T_2^*(x) &= 8x^2 - 8x + 1 \\ T_3^*(x) &= 32x^3 - 48x^2 + 18x - 1 \\ T_4^*(x) &= 128x^4 - 256x^3 + 160x^2 - 32x + 1 \\ &\dots \end{aligned}$$

and

$$\begin{aligned} 1 &= T_0^* \\ x &= \frac{1}{2}(T_0^* + T_1^*) \\ x^2 &= \frac{1}{8}(3T_0^* + 4T_1^* + T_2^*) \\ x^3 &= \frac{1}{32}(10T_0^* + 15T_1^* + 6T_2^* + T_3^*) \\ x^4 &= \frac{1}{128}(35T_0^* + 56T_1^* + 28T_2^* + 8T_3^* + T_4^*) \\ &\dots \end{aligned}$$

More extensive tables are given in the National Bureau of Standards.¹
The recurrence relation for the shifted polynomials is

$$T_{n+1}^*(x) = (4x - 2)T_n^*(x) - T_{n-1}^*(x) \quad T_0^* = 1$$

or

or

$$x T_n^*(x) = \frac{1}{4} T_{n+1}^*(x) + \frac{1}{2} T_n^*(x) + \frac{1}{4} T_{n-1}^*(x)$$

where

$$T_n^*(x) = \cos[n \arccos(2x - 1)] = T_n(2x - 1)$$

¹ Tables of Chebyshev Polynomials $S_n(x)$ and $C_n(x)$, *Natl. Bur. Std. (U.S.), Appl. Math. Ser. 9*, 1952.

PROBLEMS

- 28.7.1 Discuss the conversion from polynomials to shifted Chebyshev polynomials and back, corresponding to Sec. 28.3.
Find the formulas corresponding to
- 28.7.2 Prob. 28.6.3
- 28.7.3 Prob. 26.8.1
- 28.7.4 Differentiation
- 28.7.5 Integration
- 28.7.6 $T_m^*(x)T_n^*(x)$

CHEBYSHEV APPROXIMATION : PRACTICE

29.1 ECONOMIZATION

If a function can be represented in the form of a rapidly converging expansion in Chebyshev polynomials, and if we truncate the expansion after the term $T_{N-1}(x)$, then the error will closely resemble the first term neglected, namely, $a_N T_N(x)$. It is a folklore belief that Chebyshev expansions are among the most rapidly converging expansions there are for functions. This is clearly false in isolated cases; for example, consider an expansion in Bessel functions with coefficients of the expansion being, say, 3, 2, 1, 0, 0, 0, 0, 0, If this function were expanded in Chebyshev polynomials, then arbitrarily far out there would be nonzero coefficients. On the other hand, an examination of the process that we gave (Sec. 28.3) for going from a Taylor series

$$f(x) = \sum_{n=0}^N a_n x^n$$

to a Chebyshev expansion

$$f(x) = \sum_{n=0}^N b_n T_n(x)$$

has

$$\sum_0^N a_n = \sum_0^N b_n$$

but

$$b_N = \frac{a_N}{2^{N-1}}$$

and

$$b_{N-1} = \frac{a_{N-1}}{2^{N-2}}$$

so we might *expect* the Chebyshev series to converge more rapidly than the Taylor series.

Lanczos¹ observed that if we expand a function in a series of Chebyshev polynomials, and if we truncate the series, then the remainder will look a lot like the first $T_N(x)$ that is neglected. He further observed that the error E committed in dropping the terms beyond $T_{n-1}(x)$ is bounded by

$$|E| \leq \sum_{k=N}^{\infty} |b_k|$$

He called this process “economization.”

Economization or processes closely allied to it are widely used in numerical methods because we tend to prefer a minimax bound to a least-squares (or any other) bound. The Chebyshev equal-ripple error is often closely approximated by the error of the economized representation, and both have a sum of squares of the error that is by definition at least as great as the least-squares fit (within roundoff, of course). On the other hand, the Chebyshev error of the least-squares fit is greater than that of the Chebyshev fit (again by definition). How do the respective errors compare? It can be shown that in a certain sense the Chebyshev approximation has a sum of squares proportional to the sum of squares of the least squares fit. On the other hand, the least-squares fit has a Chebyshev error that is a multiplier of the Chebyshev minimum fit and grows with N , the degree of the fit. Thus as a general rule when in doubt about whether to use a least-squares or a Chebyshev fit: if you choose the Chebyshev fit and are wrong, then the error is a fixed multiplier (somewhat larger than 1) times the least-squares error; but if you choose the least squares and should have picked a Chebyshev,

¹ Cornelius Lanczos (1893–).

then you can commit a large error, and the higher the order of the approximating polynomial, the (probably) higher percentage error you will make. To see why this is so, you need only to look at the typical least-squares fit. Assuming that the error is very like the first orthogonal polynomial dropped, say the Legendre to keep the two cases comparable, we see from Fig. 26.7.1 that the polynomial shoots up at the ends of the interval very rapidly, thus producing the large Chebyshev error. The Chebyshev polynomials on the other hand are reasonably close to the orthogonal polynomials over most of the interval.

The field of Chebyshev approximation is now highly developed, and we refer the reader to textbooks on the subject by Fox and Parker [14] and by Snyder [53].

The obvious use for Chebyshev approximation is in the library routines for evaluating the special functions such as sine, cosine, logarithm, exponential, etc., where customarily the routines are uniformly accurate (Chebyshev) throughout the range (as measured by relative accuracy).

Another use that seldom seems to occur to users is in a problem where there is a block of heavy computation with one number as input and one number as output. Clearly this can be viewed as a table, and a Chebyshev approximation will often greatly speed up the evaluation. Indeed, in real time problems it may be the only way to save enough time to complete a full cycle on time.

29.2 ON FINDING A CHEBYSHEV EXPANSION (ECONOMIZATION)

Evidently the problem of finding a Chebyshev expansion is central to finding a Chebyshev approximation. While the error in the economization process may not be exactly equal-ripple, it will generally be very close, and in any case it will furnish a very good first approximation (Sec. 29.10).

In Sec. 28.3 we showed how to go from a Taylor series of a corresponding Chebyshev series *provided* the Taylor series was truncated at some place for which the error caused by the dropped terms was negligible. As an example of this process of conversion to Chebyshev and subsequent economization, consider the following trivial example (trivial so that it can be followed easily). Suppose we use

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} \quad (\text{approximately})$$

We will consider the interval $0 \leq x \leq 1$, and so we use the shifted polynomials $T_k^*(x)$ (Sec. 28.7). We have

$$\begin{aligned} x &= \frac{1}{2}(T_0^* + T_1^*) \\ -\frac{x^2}{2} &= -\frac{1}{16}(3T_0^* + 4T_1^* + T_2^*) \\ \frac{x^3}{3} &= \frac{1}{96}(10T_0^* + 15T_1^* + 6T_2^* + T_3^*) \\ -\frac{x^4}{4} &= \frac{-1}{512}(35T_0^* + 56T_1^* + 28T_2^* + 8T_3^* + T_4^*) \end{aligned}$$

or

$$\ln(1+x) = \frac{535}{1,536}T_0^* + \frac{19}{64}T_1^* - \frac{7}{128}T_2^* - \frac{1}{192}T_3^* - \frac{1}{512}T_4^*$$

In this approximation we know that $|T_k^*(x)| \leq 1$ ($0 \leq x \leq 1$), and so if we drop both $T_3^*(x)$ and $T_4^*(x)$, the additional error is bounded by

$$\frac{1}{192} + \frac{1}{512} = \frac{11}{1,536}$$

Indeed, also dropping the $T_2^*(x)$ would leave the additional error less than the error (1/3) due to dropping the cubic in the power-series approximation.

We see that the additional error due to the economization is indeed much as we said it would be. Thus this example, although unrealistic, shows the desired effects.

29.3 THE DIRECT EVALUATION OF THE COEFFICIENTS

The orthogonality relations enable us to compute directly the coefficients of a Chebyshev polynomial expansion,

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k T_k(x)$$

We know that

$$a_k = \frac{2}{\pi} \int_{-1}^1 f(x) \frac{T_k(x)}{\sqrt{1-x^2}} dx$$

The change of variable to θ makes the relations into

$$a_k = \frac{2}{\pi} \int_0^\pi f(\cos \theta) \cos k\theta d\theta$$

and these can be evaluated by a variant of the Fast Fourier Transform (FFT) (Chap. 33).

For example, consider finding the Chebyshev polynomial expansion of e^{-x} . We have from the definition

$$\frac{2}{\pi} \int_0^\pi e^{-t \cos \theta} \cos k\theta \, d\theta = 2I_k(t)$$

of the modified Bessel function of order k and argument t that if we put $t = 1$, we have $a_k = 2I_k(1)$, and

$$e^{-x} = I_0(1) + 2 \sum_{k=1}^{\infty} I_k(1) T_k(x)$$

Even if we do not know the analytic form of the integrals, for some functions they can be evaluated numerically if necessary.

For the Chebyshev expansion in terms of a discrete set of points x_i , or t_i , we have

$$f(x) = \frac{A_0}{2} + \sum_{k=1}^{\infty} A_k T_k(x)$$

and using, say, t_j ,

$$A_k = \frac{2}{N} \sum_{j=1}^N f(t_j) T_k(t_j)$$

What is the relation between the coefficients A_k of the discrete expansion and the a_k of the continuous expansion? The aliasing discussed in Sec. 28.4 (and again in Sec. 31.6) shows that if we have for the continuous expansion

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k T_k(x)$$

and compute the coefficients of the discrete expansion

$$\begin{aligned} A_m &= \frac{1}{N} \sum_{j=1}^N f(x_j) T_m(x_j) \\ &= \frac{1}{N} \sum_{j=1}^N \left[\frac{a_0}{2} T_m(x_j) + \sum_{k=1}^{\infty} a_k T_k(x_j) T_m(x_j) \right] \end{aligned}$$

we get

$$A_m = a_m - a_{2N-m} - a_{2N+m} + a_{4N-m} + a_{4N+m} + \cdots, \quad m = 1, 2, \dots, N-1$$

$$\frac{A_0}{2} = \frac{a_0}{2} - a_{2N} + a_{4N} - a_{6N} + \cdots \quad m = 0$$

Note that the formula for A_0 agrees with the one for the general m .

If the series converges rapidly, then this aliasing effect is small; if the convergence is slow, then the finite approximation may be far from the continuous one. For rapidly converging series the finite sums give ideal formulas for computing the integrals that arise in the continuous case.

29.4 A DIRECT METHOD

Another method for finding a Chebyshev expansion so that the error resembles $T_N(x)$ is to evaluate the function at the zeros x_i of

$$T_N(x_i) = 0$$

and then simply find the polynomial through these points, using perhaps divided differences (Secs. 18.2 and 18.3). The method is obvious and need not be discussed further.

29.5 THE CHEBYSHEV EXPANSION OF AN INTEGRAL

In the next case suppose we wish to represent some integral

$$\int_a^x f(t) dt = \sum_{k=0}^{\infty} a_k T_k(x)$$

To be specific, suppose we try

$$\arctan x = \int_0^x \frac{dt}{1+t^2}$$

It is awkward to differentiate a Chebyshev series, but it is easy to integrate. So we begin by writing

$$\frac{1}{1+x^2} = \sum_{k=0}^{\infty} b_k T_k(x)$$

But it is clear from symmetry that only b_{2k} terms will appear ($b_{2k+1} = 0$), and so we write (see Prob. 28.6.2)

$$\begin{aligned} \frac{1}{1+x^2} &= \sum_{k=0}^{\infty} b_{2k} T_{2k}(x) \\ 1 &= (1+x^2) \sum_{k=0}^{\infty} b_{2k} T_{2k}(x) \\ &= \sum_{k=0}^{\infty} b_{2k} T_{2k}(x) + b_0 x^2 T_0 + \sum_{k=1}^{\infty} b_{2k} x^2 T_{2k}(x) \\ &= \sum_{k=0}^{\infty} b_{2k} T_{2k}(x) + \frac{b_0}{2} [T_0 + T_2] + \sum_{k=1}^{\infty} \frac{b_{2k}}{4} (T_{2k-2} + 2T_{2k} + T_{2k+2}) \end{aligned}$$

Since the $T_k(x)$ are linearly independent, we may equate coefficients. The first few equations are

$$T_0: \quad 1 = b_0 + \frac{b_0}{2} + \frac{b_2}{4}$$

$$T_2: \quad 0 = b_2 + \frac{b_0}{2} + \frac{b_2}{2} + \frac{b_4}{4}$$

$$T_4: \quad 0 = b_4 + \frac{b_2}{4} + \frac{b_4}{2} + \frac{b_6}{4}$$

$$T_6: \quad 0 = b_6 + \frac{b_4}{4} + \frac{b_6}{2} + \frac{b_8}{4}$$

Except for the first two equations, the rest have the form

$$\frac{b_{2(k-1)}}{4} + \frac{3}{2}b_{2k} + \frac{b_{2(k+1)}}{4} = 0$$

If we can satisfy these and have two parameters left, we can (probably) handle the first two equations. These difference equations have the characteristic equation

$$\rho^2 + 6\rho + 1 = 0$$

or

$$\rho_1 = -3 + 2\sqrt{2} \approx -0.172$$

$$\rho_2 = -3 - 2\sqrt{2} \approx -5.828$$

Hence

$$b_{2k} = C_1(\rho_1)^k + C_2(\rho_2)^k$$

Unless $C_2 = 0$, the original series will diverge; thus we have

$$b_{2k} = C_1(\rho_1)^k \quad k = 1, 2, 3, \dots$$

We have left the two equations

$$1 = \frac{3b_0}{2} + \frac{b_2}{4}$$

$$0 = \frac{b_0}{2} + \frac{3b_2}{2} + \frac{b_4}{4}$$

with b_0 and C_1 as parameters. In more detail

$$4 = 6b_0 + C_1\rho_1$$

$$0 = 2b_0 + (6\rho_1 + \rho_1^2)C_1$$

Thus we can find b_0 and C_1 and determine the series for $1/(1+x^2)$. But $\rho^2 + 6\rho = -1$. It is now easy to integrate this term by term (Sec. 28.6) to get the Chebyshev expansion of $\arctan x$.

29.6 LANCZOS' τ PROCESS

The idea behind the τ process introduced by Lanczos is that instead of approximately solving the given problem, we can perturb it slightly and solve the approximate problem exactly. While the two processes sound different, they come down to the same thing in the long run. Still, we shall start with Lanczos' approach and later take up the more direct process.

Given, say, a differential equation

$$y' + y = 0 \quad y(0) = 1 \quad 0 \leq x \leq 1$$

which we cannot solve in a polynomial form (the solution being e^{-x}), consider, instead, solving the problem

$$y' + y = \tau T_n^*(x)$$

where τ is to be found. We see that if τ is small, then we have perturbed the differential equation only slightly. To keep the computation easy to follow, we set $n = 4$ and try

$$y = a + bx + cx^2 + dx^3 + ex^4$$

We get, equating coefficients of the like power of x ,

$$\begin{aligned} b + a &= \tau \\ 2c + b &= -32\tau \\ 3d + c &= 160\tau \\ 4e + d &= -256\tau \\ e &= 128\tau \end{aligned}$$

The initial condition fixes $a = 1$. Hence, in turn, we obtain

$$\begin{aligned} b &= \tau - 1 \\ c &= \frac{1 - 33\tau}{2} \\ d &= \frac{1}{3} \left(160\tau + \frac{33\tau - 1}{2} \right) = \frac{353\tau - 1}{6} \\ e &= \frac{1}{4} \left(-256\tau + \frac{1 - 353\tau}{6} \right) = \frac{1}{24} (1 - 1,889\tau) \\ e &= 128\tau = \frac{1}{24} (1 - 1,889\tau) \\ \tau &= \frac{1}{4,961} \end{aligned}$$

Thus the error that we have put in the original equation is

$$\left| \frac{T_4^*(x)}{4,961} \right| \leq \frac{1}{4,961}$$

and for this equation we have an exact solution

$$y = 1 - \frac{4,960}{4,961}x + \frac{2,464}{4,961}x^2 - \frac{768}{4,961}x^3 + \frac{128}{4,961}x^4$$

as compared with the series solution

$$y = 1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24}$$

whose maximum error is about

$$\frac{1}{5!} = \frac{1}{120}$$

In this case it is easy to compute a bound on the error due to the addition of the Chebyshev term to the differential equation

$$y' + y = \tau T_4^*(x)$$

$$y(x) = e^{-x} \cdot 1 + \tau e^{-x} \int_0^x T_4^*(\theta) e^\theta d\theta$$

For $0 \leq x \leq 1$

$$\begin{aligned} |y - e^{-x}| &= \left| \tau e^{-x} \int_0^x T_4^*(\theta) e^\theta d\theta \right| \\ &\leq \tau e^{-x} \int_0^x e^\theta d\theta = \tau(1 - e^{-x}) \leq \frac{0.665}{4,961} \approx 1.34 \times 10^{-4} \end{aligned}$$

We have illustrated the τ method with a simple example. Lanczos' book [33] gives a much more extensive discussion of the method. However, the main idea is the same; we slightly alter the problem and obtain an exact solution of the new problem. Since the alteration is in the domain of the physics of the original problem, we can usually understand the change more readily than some approximation made in the middle of some numerical computation. The τ method illustrates the importance of carefully answering the fourth basic question, What accuracy?

PROBLEMS

29.6.1 Apply the τ method to

$$xy'' + y = 0 \quad \begin{cases} y(0) = 0 \\ y'(0) = 1 \end{cases}$$

for $0 \leq x \leq 1$, using $n = 4$.

29.7 THE DIRECT METHOD FOR DIFFERENTIAL EQUATIONS

Often the function we want to expand in a Chebyshev series is defined by a differential equation. Again this can often be used to get the expansion directly. For example, again suppose we have the simple differential equation

$$y' = -y \quad y(0) = 1 \quad -1 \leq x \leq 1$$

It is usually preferable to suppose that we have the expansion of the derivative of the function

$$y' = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k T_k$$

From Sec. 28.6 we can integrate

$$y(x) = \frac{a_0 T_1}{2} + a_1 \frac{T_2}{4} + \sum_{k=2}^{\infty} \frac{a_k}{2} \left[\frac{T_{k+1}}{k+1} - \frac{T_{k-1}}{k-1} \right] + C$$

We equate coefficients

$$T_0: \quad \frac{-a_0}{2} = C$$

$$T_1: \quad -a_1 = \frac{a_0}{2} - \frac{a_2}{2}$$

$$T_2: \quad -a_2 = \frac{a_1}{4} - \frac{a_3}{4}$$

$$T_3: \quad -a_3 = \frac{a_2}{6} - \frac{a_4}{6}$$

$$T_4: \quad -a_4 = \frac{a_3}{8} - \frac{a_5}{8}$$

The general difference equation for the coefficients is

$$a_{k+1} - 2ka_k - a_{k-1} = 0$$

and has two solutions, one growing with k and one decaying. From Sec. 13.8 we see that we want to start at the top end and solve downward. If, as before, we want only through T_4 , we set $a_5 = 0$ and tentatively $a_4 = 1$. As the result of computing step by step

$$a_3 = -8a_4$$

$$a_2 = 6\left(\frac{a_4}{6} - a_3\right)$$

$$a_1 = 4\left(\frac{a_3}{4} - a_2\right)$$

$$a_0 = \left(\frac{a_2}{2} - a_1\right)$$

we will find that usually at $x = 0$

$$a_0 T_0 + a_2 T_2 + a_4 T_4 \neq 1$$

but that by making the obvious scale factor change an a_4 (and also a_3, a_2, a_1, a_0), we will satisfy the initial condition.

An examination of this process will reveal that it is equivalent to the τ process of rigging things so that the final equations are satisfied, since there we added the term we needed to produce an exact cancellation for the last power of x we used.

29.8 THE EVALUATION OF CHEBYSHEV EXPANSIONS

A Chebyshev polynomial expansion may be evaluated in a direct fashion. The method is based on the three-term recurrence relation

$$T_{k+1} = 2xT_k - T_{k-1}$$

If, in some fashion, we had the first two values, then each iteration would produce the next Chebyshev polynomial.

Let us examine the process more closely. Suppose we have three locations A , B , and C . What we do is multiply the contents of B by $2x$ and from this subtract the contents of C , and put this result in A . We then shift B to C and A to B . Imagine that x is fixed and that B and C both have zeros in them. Then each iteration produces more zeros. But if we were to add to B the quantity x and to C the quantity 1, then we would start to generate the Chebyshev polynomials step by step, and after k steps we would have $T_{k+1}(x)$.

If we put in $a_N x$ and a_N instead of x and 1, we would have after $N - 1$ iterations exactly $a_N T_N(x)$. If after the first iteration we added $a_{N-1}x$ and a_{N-1} in B and C , we would end up with the sum

$$a_N T_N(x) + a_{N-1} T_{N-1}(x)$$

And in general, if at each step we added in the two numbers corresponding to the term a_k that was appropriate, we would end up with the sum of the Chebyshev series. Often this process is preferable to converting to a polynomial in x and then evaluating the result. The roundoff effect in the polynomial case may be more severe than in the direct evaluation.

29.9 THROWBACK

The aliasing of the higher polynomials at the zeros of $T_N(x)$ suggests that instead of throwing away all the higher terms of a Chebyshev expansion, as is done in the economization process, they should be “thrown back” (Sec. 18.8) via the aliasing

relations. Thus while we still cannot be sure that the error curve is equal-ripple, we can be sure that the error curve has the proper zeros when we drop the $T_N(x)$ term. We will not pursue this further.

29.10 LEVELING THE ERROR CURVE

We have seen that we can find a Chebyshev polynomial expansion for many functions, but that when we economize it we do not get the exact equal-ripple error curve.

There are two broad approaches, with many refinements to each. The first looks at the positions of the zeros of the error curve and the values of the maxima between them and at the ends. In view of these maxima the zeros are moved a bit to form the next approximation. Evidently if a maximum or minimum is larger than the others, the adjacent (or nearest, if it is in an end interval) zeros are moved a bit together under the reasonable assumption that this will decrease the size of the peak. Each cycle consists of carefully computing the error curve and finding the size and approximate location of the extremes. From this the motion of the zeros is found. Usually a few iterations suffice to level the curve at least by a factor of 10.

The second approach is more mathematical. Having again located the extremes, both amount and location, the coefficients are perturbed one at a time to see how they alter the extremes (in the Chebyshev polynomial case these amounts can be found from the polynomials themselves). Then a set of simultaneous linear equations are set up with arbitrary changes in the coefficients so that the result (if it were all linear) will exactly annihilate the inequalities of the peaks. Repeated trials will again level the curve (most of the time).

Another approach to leveling the error curve is to use the sigma factors of Sec. 32.7.

*RATIONAL FUNCTION APPROXIMATION

30.1 INTRODUCTION

Up to this point in Part II we have assumed that between the sample points the function behaved as a polynomial. In this chapter we shall assume that the function behaves as the quotient of two polynomials, which is usually called a *rational function*. Rational functions have the two fundamental properties of still being a rational function under both translation and scale change of the independent variable.

Rational functions have a number of properties that are frequently needed for an approximation leading to the analytic substitution of a new function in place of the given one. Most noticeable is the fact that rational functions can approximate functions having infinite values y_i for finite values x_i . Another feature is that they can approximate straight lines, especially the x axis, for large values of x_i , which is something that nontrivial polynomials cannot do.

Inside a computer, rational functions are rapid and easy to evaluate; hence they are often used to provide a computable approximation to some of the more difficult functions. For such purposes a Chebyshev type of minimax fit is likely to be needed.

The theory of approximation by rational functions is in a confused, rapidly evolving state, and we shall give only the simplest approach. One reason for not going farther is that in computing, the average person is not likely to do many rational approximations in a year, and more efficient methods, which save a few seconds on a fast computer, do not seem to be worth the time in an elementary course.

30.2 THE DIRECT APPROACH

Suppose we have either a function from which we can compute some samples (x_i, y_i) or the samples themselves, and we wish to fit a rational function to the data. Thus we want

$$y = f(x) \approx \frac{N(x)}{D(x)}$$

where $N(x)$ and $D(x)$ are polynomials.

The first step is to decide which polynomial forms to try. In one particular case, plotting

$$\log y_i \text{ versus } \log x_i$$

for small $x_i > 0$ produced a slope of approximately 3, from which we deduced that we should try

$$y = f(x) \simeq \frac{ax^3 + \cdots}{1 + \cdots}$$

For large x the original data fell on what appeared to be a straight line of positive slope. This determined that the degree of $N(x)$ should be 1 greater than the degree of $D(x)$. Let the degree of $D(x)$ be k . Then

$$y = f(x) = \frac{a_3 x^3 + a_4 x^4 + \cdots + a_{k+1} x^{k+1}}{1 + b_1 x + b_2 x^2 + \cdots + b_k x^k} \quad (30.2.1)$$

It remains to choose k . It is hard to say just how to choose k , except by noting the number of parameters that will be available and by comparing that (subjectively) with the "complexity" of the data.

When k is chosen, we then know the number of parameters M . Note that we must fix one coefficient because only the ratio of $N(x)$ and $D(x)$ occurs; hence one coefficient (known to be nonzero) must be chosen as a fixed number.¹

¹ In practice, it is usually best to select 1 as the highest power either of the numerator or of the denominator, since this saves one multiplication when evaluating the rational function.

We either select or compute the appropriate number M of samples, located where they seem to be most important. We then have

$$D(x_i)y_i = N(x_i) \quad i = 1, \dots, M$$

as a set of M equations to be solved. In the example we have from Eq. (30.2.1)

$$\begin{aligned} y_i + b_1 x_i y_i + b_2 x_i^2 y_i + \dots + b_k x_i^k y_i \\ = a_3 x_i^3 + a_4 x_i^4 + \dots + a_{k+1} x_i^{k+1} \end{aligned} \quad (30.2.2)$$

for $i = 1, 2, \dots, 2k - 1$.

These are linear equations in the unknowns a_i and b_i and can be solved by a library routine.

The point is often raised that such a system is of high degree. As far as can be determined, seldom does the number of unknowns exceed 10 or 12, and thus the solution involves no more than a few thousand operations; this is hardly serious on a fast machine, if it is to be done once in a problem.

30.3 LEAST-SQUARES FITTING BY RATIONAL FUNCTIONS

Often there are more data points (x_i, y_i) ($i = 1, 2, \dots, M$) than parameters in the rational function we wish to fit. In such cases one approach is to find a least-squares fit. We want, therefore, to find the coefficients a_i and b_i such that

$$\min = \sum_{i=1}^M \left[y(x_i) - \frac{N(x_i)}{D(x_i)} \right]^2$$

The unknown coefficients occur nonlinearly in the normal equations, and this makes them difficult to solve. One way around this is to note that we can write the expression we wish to minimize in the form

$$\min = \sum_{i=1}^M \frac{1}{D^2(x_i)} [D(x_i)y(x_i) - N(x_i)]^2$$

We now set about finding an iterative solution to the following problem:

$$\min_k = \sum_{i=1}^M \frac{1}{D_{k-1}^2(x_i)} [D_k(x_i)y(x_i) - N(x_i)]^2 \quad k = 1, 2, \dots$$

where we take, if nothing else is known, $D_0 \equiv 1$. The unknowns now occur linearly in the normal equations.

In the usual course of events the convergence is rapid. However, as is always possible, the iterative loop with the feedback may cause an instability. If

instability occurs, an examination of the nature of the trouble together with commonsense, will usually produce a method for dampening the instability; hence the elaborate theory of the general case is not worth going into here.

30.4 CHEBYSHEV APPROXIMATION BY RATIONAL FUNCTIONS

As indicated in Sec. 30.1, frequently a rational function is used as an easily computable approximation to a transcendental function. When such an approximation is to be used many times, either in some frequently recurring problem or in a special-purpose computer, then a Chebyshev equal-ripple (or minimax) approximation is likely to be desired.

We begin by choosing a set of samples x_i , perhaps the zeros of a suitable $T_k(x)$, and we compute the corresponding y_i . From these we find the corresponding rational function of the appropriate form (as in Sec. 30.2).

We then plot the error curve by computing the error at many points. For intervals where the error is large, we want to move the sample points closer together, and where the local extreme error is small, we want to move the samples farther apart. With this new choice of sample points, we repeat the process. Seldom are as many as 10 trials necessary to come quite close to an equal-ripple error curve. The speed of the approach to the equal-ripple error curve depends, among other things, on the method used for respacing the samples.

We offer here no proof that this heuristic method will work, but experience indicates that a number of such simple schemes work reasonably well. A little "low cunning" in arranging the computation is usually necessary to avoid the loss of too many figures.

30.5 RECIPROCAL DIFFERENCES

In Secs. 30.3 and 30.4 we used iterative schemes which require a number of trials, each of which involves the solution of the system of linear equations. This suggests that we examine the possibilities of systematizing the solution.

Following Milne [38], we consider the special case

$$y = \frac{a_0 + a_1x + a_2x^2 + a_3x^3}{b_0 + b_1x + b_2x^2}$$

which uses six data points (x_i, y_i) ($i = 1, \dots, 6$). Multiplying, we get

$$a_0 - b_0y + a_1x - b_1xy + a_2x^2 - b_2x^2y + a_3x^3 = 0$$

which is of the same form as before except that it is of sixth order.

We repeat the process. Multiply:

- 1 Column 1 by $\rho_1(x_2, x_1)$ and subtract from column 2
- 2 Column 3 by $\rho_1(x_2, x_1)$ and subtract from column 4
- 3 Column 5 by $\rho_1(x_2, x_1)$ and subtract from column 6
- 4 Column 3 by x_2 and subtract from column 5
- 5 Column 1 by x_2 and subtract from column 3

Reduce the determinant and divide:

- 6 Row 1 by $\rho_1(x, x_1) - \rho_1(x_2, x_1)$
- 7 Row 2 by $\rho_1(x_3, x_1) - \rho_1(x_3, x_1)$

.....

Again a wise choice of the row in which we produce 1, 0, 0, 0, 0 can help to keep accuracy. If we write

$$u_x = \frac{x - x_2}{\rho_1(x, x_1) - \rho_1(x_2, x_1)}$$

we get

$$\begin{vmatrix} 1 & u_x & x & xu_x & x^2 \\ 1 & u_3 & x_3 & x_3 u_3 & x_3^2 \\ 1 & u_4 & x_4 & x_4 u_4 & x_4^2 \\ 1 & u_5 & x_5 & x_5 u_5 & x_5^2 \\ 1 & u_6 & x_6 & x_6 u_6 & x_6^2 \end{vmatrix} = 0$$

The quantity $\rho_1(x_i, x_j)$ was clearly symmetric in its variables. But this is not true of the quantities that we have labeled u_i . It is customary to use the symmetric quantities

$$\begin{aligned} & \frac{x - x_2}{(x - x_1)/(y - y_1) - (x_2 - x_1)/(x_2 - y_1)} + y_1 \\ & \equiv \frac{x - x_1}{(x - x_1)/(y - y_1) - (x_2 - x_1)/(y_2 - y_1)} + y_1 \\ & \equiv \frac{x_2 - x_1}{(x_2 - x)/(y_2 - y) - (x - x_1)/(y - y_1)} + y \end{aligned}$$

and call them

$$\rho_2(x, x_1, x_2)$$

but the symmetry is of little value in practice. Continuing this process, we reduce the determinant to second order which is then trivial to handle.

We shall not go further in this highly developed field but merely refer the reader to standard books such as Milne [38], Milne-Thomson [40], and Ralston [49].

PART III

Fourier Approximation— Modern Theory

31.1 ORIENTATION

Part I was concerned with the fundamentals of computing and a number of various algorithms. Part II concentrated on approximating infinite operators like interpolation, integration, and differentiation by using polynomials. Numerous formulas were found, but the essential content of Part II was the methodology of finding formulas, as well as the rich variety of possible formulas. In Part III we use this methodology, but we use a different class of approximation functions; instead of polynomials we use sums of sines and cosines. In Chaps. 31 to 33 in Part III we use discrete sums of these functions, in the form of a Fourier series, to approximate periodic functions, while later we use a continuous, noncountable number of the functions, in the form of a Fourier integral, to approximate more general functions.

Because the Fourier series and integral *tend* to be ignored in modern mathematics (though they gave rise to a great deal of mathematics), we have to develop the mathematical manipulations we need. To keep the material within reasonable bounds the more formal mathematical rigor will be ignored [34, 35, 64].

It is recommended that Sec. 1.6 be reread at this point, and possibly all of Chap. 1, so that the material discussed in Part III will be seen in reasonable perspective.

By using Fourier approximation instead of polynomial approximation we avoid the many troubles of polynomial approximation that we have frequently mentioned. Not only does the Fourier model provide an alternative technique for finding formulas, and hence sometimes different formulas, but it also gives an alternate way of interpreting the same formulas found in Part II. Chapter 35 re-examines many of the results obtained in Part II and shows how the alternate view can often, but not always, shed a great deal of light on what the answers mean. Thus we have a new tool for fulfilling the purpose of the book as expressed in its motto, *The Purpose of Computing Is Insight, Not Numbers*.¹

The frequency approach of Part III is very powerful in the planning stage, because it often enables us to make very accurate estimates of the spacing of the samples necessary to obtain a given accuracy. In the computing stage, it likewise offers not only many of the old formulas, but also others that are essentially new and which may well meet the needs of the problem better. Probably most important, in the interpretation of the results back to the physical problem, the frequency approach can interpret in a new light not only the numbers, but also the details of how the computation went, such as the physical meaning of the tight or loose spacing of the samples. Finally, it often provides very powerful ways of verifying the physical model in many of its aspects, thus reassuring the proposer of its validity and shedding new light on the original source of the computing. We can therefore put into practical use the theory we have learned.

The author has wondered for more than 20 years why the textbooks and mathematically inclined computer journals have almost totally neglected this fruitful and valuable approach to computing. The answers seem to be (1) that the Fourier series and integral are seldom taught in mathematical circles except as special cases of generalizations that rob them of any physical meaning, and this ignorance tends to be followed by contempt for the knowledge; and (2) the Fourier approach puts much emphasis on the connection between the problem and the method used, and therefore the method cannot be studied in the vacuum of pure mathematics. As a result of this neglect the reader who wishes to follow up on this approach and find out what is new is referred to the engineering journals like the *IEEE Transactions on Computers* and the *Transactions on Automatic Control*,² plus textbooks on topics such as automatic control [54] and digital filters [32].

¹ It is sometimes suggested that the motto be revised to, *The Purpose of Computing Numbers Is Not Yet in Sight*.

² IEEE, 345 East 47th Street, New York, N.Y., 10017.

31.2 THE EFFECT OF SAMPLING—ALIASING

Most computation is done with equally spaced samples. Let this spacing be chosen as the unit (of time); that is, $h = \Delta t = 1$. Section 1.6 gave one description of this effect, namely that due to the sampling, some forward rotations will appear as backward rotations. A practical application of this effect is the stroboscope which flashes a light at slightly less than the period of some rotating equipment being viewed. As a result the equipment appears to be rotating at a very slow rate; the closer the two rates, the slower the appearance of rotation.

This effect can be understood from a simple trigonometric identity. Corresponding to the sinusoid

$$\cos [\pi(n + \epsilon)t + \phi]$$

there is another sinusoid

$$\cos [\pi(n - \epsilon)t - \phi]$$

which has the same sample values at the equally spaced sample points as can be seen in Fig. 31.2.1. The equality of the sample values follows from

$$\cos[\pi(n + \epsilon)t + \phi] - \cos[\pi(n - \epsilon)t + \phi] = -2 \sin(n\pi t) \sin(\pi \epsilon t + \phi) = 0$$

where both n and t are integers. Thus the sampling process confuses the two different frequencies $\pi(n + \epsilon)$ and $\pi(n - \epsilon)$ —both appear as if they were the same frequency.

It is customary to refer to this confusion as aliasing,¹ that is, various frequencies adopting the name of one particular frequency, which is usually chosen to be the lowest positive frequency of all those which are aliased. It is obvious that *aliasing is a consequence of sampling at equally spaced intervals and cannot be undone once the sampling occurs.*

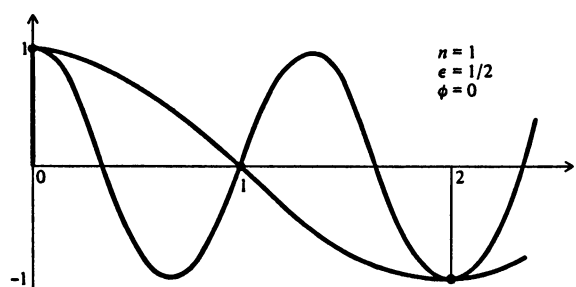


FIGURE 31.2.1

¹ Apparently introduced by J. W. Tukey (1915-).

In a sense, the aliasing of the Chebyshev polynomials in Sec. 28.4 was that of the cosines in the disguise of

$$x = \cos \theta$$

More generally, aliasing also occurs in polynomial sampling, though it is not usually recognized as such. Consider polynomial approximation based on n sample points x_i ($i = 1, 2, \dots, n$). The polynomial

$$\pi(x) = (x - x_1)(x - x_2) \cdots (x - x_n)$$

plays a central role in the theory. Let $P_m(x)$ be any polynomial of degree m (we may think of m as being greater than n though it need not be). When we divide $P_m(x)$ by $\pi(x)$ we get a quotient $Q(x)$ and a remainder $R(x)$, that is,

$$P_m(x) = Q(x)\pi(x) + R(x)$$

This expression is the obvious generalization of the remainder theorem, namely *at the sample points x_i ,*

$$P_m(x_i) = R(x_i) \quad (i = 1, 2, \dots, n)$$

and due to the sampling one function cannot be distinguished from the other *if* all we have are the sample values.

If we imagine doing this for each x^m ($m = 0, 1, \dots$) we have the correspondence

$$x^m \rightarrow R_m(x)$$

Now for a Taylor's series expansion of any analytic function $f(x)$

$$f(x) = \sum_{m=0}^{\infty} a_m x^m \rightarrow \sum_{m=0}^{\infty} a_m R_m(x)$$

and at the sample points

$$f(x_i) = \sum_{m=0}^{\infty} a_m R_m(x_i)$$

The summation is *the* interpolating polynomial of degree n which we would get if we passed an interpolating polynomial through the given samples (neglecting roundoff, of course).

31.3 THE CONTINUOUS FOURIER EXPANSION

In Sec. 26.2 we used the Fourier series as an example of a set of orthogonal functions. We showed by direct integration that

$$\int_0^{2\pi} \cos mx \cos nx \, dx = \begin{cases} 2\pi & m = n = 0 \\ \pi & m = n \neq 0 \\ 0 & m \neq n \end{cases}$$

$$\int_0^{2\pi} \cos mx \sin nx \, dx = 0$$

$$\int_0^{2\pi} \sin mx \sin nx \, dx = \begin{cases} 0 & m = n = 0 \\ \pi & m = n \neq 0 \\ 0 & m \neq n \end{cases}$$

and hence if

$$F(x) = \frac{a_0}{2} + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \cdots$$

then the Fourier coefficients are given by

$$a_k = \frac{1}{\pi} \int_0^{2\pi} F(x) \cos kx \, dx$$

$$b_k = \frac{1}{\pi} \int_0^{2\pi} F(x) \sin kx \, dx$$

We also showed that (1) these Fourier coefficients give the least-squares fit to the function, and (2) that then Bessel's inequality

$$\int_0^{2\pi} F^2(x) \, dx - \pi \left[\frac{a_0^2}{2} + \sum_{k=1}^N (a_k^2 + b_k^2) \right] \equiv \sum_{i=1}^M e_i^2 \geq 0$$

gives a measure of the goodness of fit.

It is well known that any reasonable engineering function (see Fig. 31.3.1)

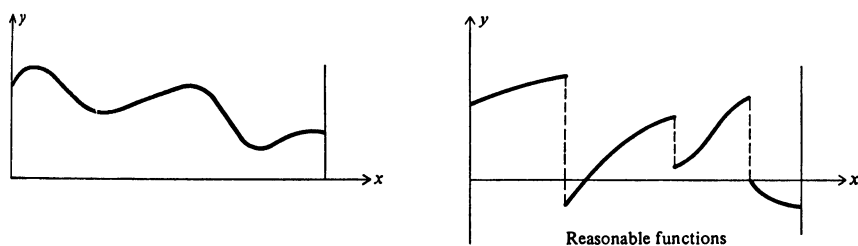


FIGURE 31.3.1

can be expanded in a Fourier series. The rate of convergence of the series depends on the discontinuities in the function and in its derivatives (see Chap 32 for more details); but we must remember that the function is assumed to be periodic and is, as it were, wrapped around a cylinder (see Fig. 31.3.2).

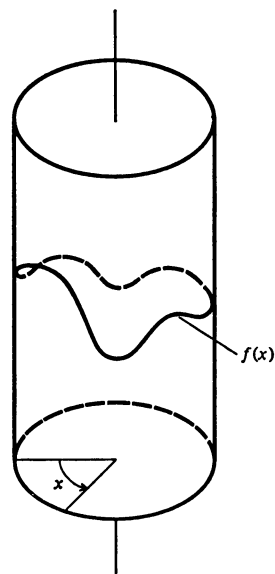


FIGURE 31.3.2

Any interval of length L may be used; in particular, it is often useful to use $-L/2 \leq x \leq L/2$ instead of $0 \leq x \leq L$.

PROBLEMS

Find the Fourier expansions for:

$$31.3.1 \quad y = x \quad -1 \leq x \leq 1$$

$$31.3.2 \quad y = 1 - x^2 \quad -1 \leq x \leq 1$$

$$31.3.3 \quad y = e^{-x} \quad 0 \leq x \leq L$$

$$31.3.4 \quad y = \begin{cases} 1 & 0 < x < \pi \\ 0 & \pi < x < 2\pi \end{cases}$$

31.4 THE COMPLEX FORM OF THE FOURIER SERIES

It is often convenient, especially for theory, to write $\sin x$ and $\cos x$ in their equivalent complex forms. Using the Euler identity

$$e^{ix} = \cos x + i \sin x$$

we replace i with $-i$

$$e^{-ix} = \cos x - i \sin x$$

and add and subtract these two equations to get

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}$$

In this notation (k an integer) we have

$$F(x) = \sum_{k=-\infty}^{\infty} c_k e^{(2\pi i/L)kx}$$

where the discrete set of c_k is found by the simple formula

$$c_k = \frac{1}{L} \int_0^L F(x) e^{-(2\pi i/L)kx} dx$$

The orthogonality condition which determines the form of the c_k is then easily proved by direct integration

$$\int_0^L e^{(2\pi i/L)kx} e^{(2\pi i/L)mx} dx = \begin{cases} 0 & m+k \neq 0 \\ L & m+k = 0 \end{cases}$$

Comparing the two forms of representation, we find

$$c_k = \begin{cases} \frac{a_k - ib_k}{2} & k > 0 \\ \frac{a_k + ib_k}{2} & k < 0 \\ \frac{a_0}{2} & k = 0 \end{cases}$$

PROBLEMS

Expand in the complex Fourier-series form:

$$31.4.1 \quad y = x \quad -\pi < x < \pi$$

$$31.4.2 \quad y = \begin{cases} 1 & 0 \leq x \leq \frac{1}{2} \\ 0 & \frac{1}{2} < x < 1 \end{cases}$$

31.5 THE FINITE FOURIER SERIES

It is a remarkable fact that the sines and cosines are orthogonal over both the continuous interval and sets of discrete, equally spaced points covering the period. *This is extremely important* because in computing we are usually given only samples of the function at a set of equally spaced points. This means, in turn, that we can find the coefficients exactly (except for roundoff) in the discrete sampled problem, instead of approximately in the continuous model by numerical integration.

We will consider only an even number $2N$ of points. Let the $2N$ sample points be (see Fig. 31.5.1)

$$x_p = 0, \frac{L}{2N}, \frac{2L}{2N}, \dots, \frac{(2N-1)L}{2N}$$

or more shortly,

$$x_p = \frac{Lp}{2N} \quad p = 0, 1, \dots, 2N-1$$

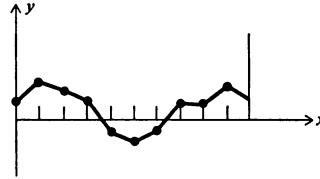


FIGURE 31.5.1

We want to show that the $2N$ Fourier functions

$$1, \cos \frac{2\pi}{L} x, \cos \frac{2\pi}{L} 2x, \dots, \cos \frac{2\pi}{L} (N-1)x, \cos \frac{2\pi}{L} Nx$$

$$\sin \frac{2\pi}{L} x, \sin \frac{2\pi}{L} 2x, \dots, \sin \frac{2\pi}{L} (N-1)x$$

form an orthogonal set of functions. This amounts to showing that for $0 \leq k$ and $m \leq N$

$$\sum_{p=0}^{2N-1} \cos\left(\frac{2\pi}{L} k \frac{Lp}{2N}\right) \cos\left(\frac{2\pi}{L} m \frac{Lp}{2N}\right) = \begin{cases} 0 & k \neq m \\ N & k = m \neq 0, N \\ 2N & k = m = 0, N \end{cases}$$

$$\sum_{p=0}^{2N-1} \cos\left(\frac{2\pi}{L} k \frac{Lp}{2N}\right) \sin\left(\frac{2\pi}{L} m \frac{Lp}{2N}\right) = 0$$

$$\sum_{p=0}^{2N-1} \sin\left(\frac{2\pi}{L} k \frac{Lp}{2N}\right) \sin\left(\frac{2\pi}{L} m \frac{Lp}{2N}\right) = \begin{cases} 0 & k \neq m \\ N & k = m \neq 0, N \\ 0 & k = m = 0, N \end{cases}$$

In fact we shall prove more, namely that when

$$k \pm m = 0, \pm 2N, \pm 4N, \dots$$

the orthogonality conditions have special features.

We begin simply by examining the series

$$\sum_{p=0}^{2N-1} e^{(2\pi i/L)kx_p} = \sum_{p=0}^{2N-1} e^{\pi i k p/N} \quad \text{for integer } k$$

which is a geometric progression with ratio

$$r = e^{\pi i k/N}$$

The sum of the geometric progression is

$$\begin{cases} \frac{1-r^{2N}}{1-r} = 0 & r \neq 1 \\ 2N & r = 1 \end{cases}$$

The upper sum is true because $r^{2N} = e^{2\pi i k} = 1$. The lower value is true because then $r = 1$. The condition $r = 1$ means $k = 0, \pm 2N, \pm 4N, \dots$

We next show that the set of functions

$$e^{(2\pi i/L)kx_p}$$

is orthogonal in the sense that the product of one function times the complex conjugate of another function summed over the points x_p is zero, that is, that

$$\sum_{p=0}^{2N-1} e^{(2\pi i/L)kx_p} e^{-(2\pi i/L)mx_p} = \begin{cases} 0 & |k-m| \neq 0, 2N, 4N, \dots \\ 2N & |k-m| = 0, 2N, 4N, \dots \end{cases}$$

This follows immediately from the above by writing the product as

$$\sum_{p=0}^{2N-1} e^{(2\pi i/L)(k-m)x_p}$$

and noting that $k-m$ plays the role of k .

We now return to the “real functions” by using the Euler identity

$$e^{ix} = \cos x + i \sin x$$

The condition for the single exponential summed over the points x_p becomes two equations (the real and imaginary parts separately)

$$\begin{aligned} \sum_{p=0}^{2N-1} \cos \frac{2\pi}{L} kx_p &= \begin{cases} 0 & k \neq 0, \pm 2N, \pm 4N, \dots \\ 2N & k = 0, \pm 2N, \pm 4N, \dots \end{cases} \\ \sum_{p=0}^{2N-1} \sin \frac{2\pi}{L} kx_p &= 0 \quad \text{for all } k \end{aligned}$$

At last we are ready to prove the orthogonality of the Fourier functions over the set of equally spaced points x_p . The first of the three orthogonality equations can be written, by using the trigonometric identity

$$\cos a \cos b = \frac{1}{2} [\cos(a + b) + \cos(a - b)]$$

$$\begin{aligned} \frac{1}{2} \sum_{p=0}^{2N-1} \left[\cos \pi(k+m) \frac{p}{N} + \cos \pi(k-m) \frac{p}{N} \right] \\ = \begin{cases} 0 & |k-m| \text{ and } |k+m| \neq 0, 2N, 4N, \dots \\ N & |k-m| \text{ or } |k+m| = 0, 2N, 4N, \dots \\ 2N & |k-m| \text{ and } |k+m| = 0, 2N, 4N, \dots \end{cases} \end{aligned}$$

For the restricted set of functions we are now using, namely,

$$\begin{aligned} 1, \cos \frac{2\pi}{L} x, \dots, \cos \frac{2\pi}{L} (N-1)x, \cos \frac{2\pi}{L} Nx \\ \sin \frac{2\pi}{L} x, \dots, \sin \frac{2\pi}{L} (N-1)x \end{aligned}$$

$k + m = 0, 2N, 4N, \dots$ cannot occur unless $k = m = 0$ or N . Thus, we have the required orthogonality of the Fourier functions over the set x_p .

The orthogonality in turn leads directly to the expansion of an arbitrary function $F(x)$ defined on the set of points x_p ,

$$F(x) = \frac{A_0}{2} + \sum_{k=1}^{N-1} \left(A_k \cos \frac{2\pi}{L} kx + B_k \sin \frac{2\pi}{L} kx \right) + \frac{A_N}{2} \cos \frac{2\pi}{L} Nx$$

where

$$\begin{aligned} A_k &= \frac{1}{N} \sum_{p=0}^{2N-1} F(x_p) \cos \frac{2\pi}{L} kx_p & k = 0, 1, \dots, N \\ B_k &= \frac{1}{N} \sum_{p=0}^{2N-1} F(x_p) \sin \frac{2\pi}{L} kx_p & k = 1, 2, \dots, N-1 \end{aligned}$$

The function $F(x)$ is often given at $2N + 1$ points, but still for $2N$ intervals, and it is assumed that

$$F(0) = F(L)$$

If this is not so, then it is customary to use

$$\frac{F(0) + F(L)}{2}$$

as the value of $F(0)$. In this case the formulas for the coefficients A_k and B_k are

$$A_k = \frac{1}{N} \left[\frac{F(0)}{2} + \sum_{p=1}^{2N-1} F(x_p) \cos \frac{2\pi}{L} kx_p + \frac{F(L)}{2} \right]$$

$$B_k = \frac{1}{N} \sum_{p=1}^{2N-1} F(x_p) \sin \frac{2\pi}{L} kx_p$$

These are effectively the trapezoid rule for numerically integrating the corresponding integrals in the continuous interval.

There is a second set of equally spaced points that we need to mention, namely,

$$t_p = x_p + \frac{L}{4N} = \frac{L}{2N} \left(p + \frac{1}{2} \right)$$

which are midway between the x_p points (recall that $x_0 = 0$ is the same as x_{2N} due to periodicity).

Repeating the steps briefly, we see that the sum

$$\sum_{p=0}^{2N-1} e^{(2\pi i k/L)t_p} = \begin{cases} 0 & r \neq 1 \\ 2N e^{\pi i k/2N} & r = 1 \end{cases}$$

as before. Hence, the rest follows as before, except for the special treatment of the end values in the summations for the coefficients.

How accurate is the finite Fourier expansion in reproducing the original data (within roundoff)? The set of functions is orthogonal; hence they are linearly independent and form a basis that spans the whole space. Thus, within roundoff, the expansion reproduces the data exactly.

PROBLEMS

Find the finite Fourier expansion for:

31.5.1 The data $x(0) = 0$; $x(1) = 1$; $x(2) = 2$; $x(3) = 3$; $N = 2$.

31.5.2 The data $x(\frac{1}{2}) = 1$; $x(\frac{3}{2}) = 1$; $x(\frac{5}{2}) = -1$; $x(\frac{7}{2}) = -1$; $N = 2$.

31.6 RELATION OF THE DISCRETE AND CONTINUOUS EXPANSIONS

It is reasonable to ask, What is the relation between the two expansions we have just found, the continuous and the discrete? Let the continuous expansion be

$$F(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} \left(a_k \cos \frac{2\pi}{L} x + b_k \sin \frac{2\pi}{L} x \right)$$

with lowercase letters for the coefficients. We pick $x_p = Lp/(2N)$ for convenience. If we multiply $F(x_p)$ by $\cos(2\pi/L)kx_p$ and sum, we get

$$\sum_{p=0}^{2N-1} F(x_p) \cos \frac{2\pi}{L} kx_p = NA_k = N(a_k + a_{2N-k} + a_{2N+k} + \cdots)$$

Hence, the coefficient we calculate is

$$A_k = a_k + \sum_{m=1}^{\infty} (a_{2Nm-k} + a_{2Nm+k})$$

which expresses the (uppercase) finite Fourier-series coefficients in terms of the (lowercase) continuous Fourier-series coefficients.

Similarly,

$$B_k = b_k + \sum_{m=1}^{\infty} (-b_{2Nm-k} + b_{2Nm+k})$$

For the special constant term A_0 , we have

$$A_0 = a_0 + 2 \sum_{m=1}^{\infty} a_{2Nm}$$

Thus, various frequencies present in the original continuous signal $F(x)$ are added together due to the sampling. This effect is called “aliasing” and is *directly attributable to the act of sampling at equally spaced points*; once the sampling has been done, the effect cannot be undone (from the samples alone).

These relationships between the coefficients of the two expansions, the continuous and the discrete sampled, show clearly the aliasing of the frequencies. If we represent the frequencies as points along a line running from zero to infinity, we can represent the aliasing by folding the line back and forth onto itself. The first frequency at which a fold occurs is called *the folding frequency*, or the *Nyquist¹ frequency*, and is $k = N$. Subsequent folds occur at this same frequency spacing. All points on the curve (shown in Fig. 31.6.1) above the same location on the

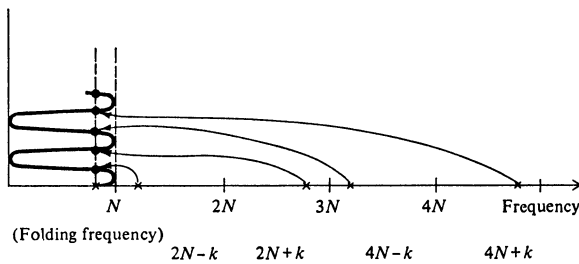


FIGURE 31.6.1
Aliasing of frequencies.

¹ Named after Harry Nyquist (1889–), a famous scientist in electrical engineering.

frequency axis appear as the same frequency owing to the effect of sampling, and once we have sampled, we cannot again separate the different frequencies that have been aliased together. At the folding frequency itself, we are sampling at the crucial rate of 2 samples per cycle [and $\sin \pi t = 0$, $\cos \pi t = (-1)^t$ where $t = \text{an integer}$].

31.7 THE POWER SPECTRUM

Up to now we have concentrated on the coefficients of the Fourier-series expansion of a periodic function. However, the coefficients themselves are often not physically significant because they depend on the choice of the origin. To see this, consider that

$$a_k = \frac{1}{\pi} \int_0^{2\pi} f(t) \cos kt \, dt$$

$$b_k = \frac{1}{\pi} \int_0^{2\pi} f(t) \sin kt \, dt$$

and under a shift of coordinates of a distance s [$f(t)$ is periodic]

$$t' = t + s$$

we would compute the primed coefficients a'_k and b'_k . For a'_k we have

$$a'_k = \frac{1}{\pi} \int_0^{2\pi} f(t+s) \cos kt \, dt$$

$$= \frac{1}{\pi} \int_s^{s+2\pi} f(t') \cos k(t'-s) \, dt'$$

Due to periodicity this is the same as

$$a'_k = \frac{1}{\pi} \int_0^{2\pi} f(t') (\cos kt' \cos ks + \sin kt' \sin ks) \, dt'$$

$$= a_k \cos ks + b_k \sin ks$$

Similarly

$$b'_k = a_k \sin ks - b_k \cos ks$$

Thus the coefficients of the expansion depend on the origin we pick for the periodic function, and usually this is of no physical significance.

However, the nonlinear expression

$$(a'_k)^2 + (b'_k)^2 = a_k^2 \cos^2 ks + 2a_k b_k \cos ks \sin ks + b_k^2 \sin^2 ks$$

$$+ a_k^2 \sin^2 ks - 2a_k b_k \cos ks \sin ks + b_k^2 \cos^2 ks = a_k^2 + b_k^2$$

The quantity $a_k^2 + b_k^2$ is invariant under translations, and is therefore much more likely to have physical significance than the individual coefficients. This quantity (or its square root) is usually called "the power at frequency k ," and a plot of

$$a_k^2 + b_k^2$$

as a function of k is called the "power spectrum."

From Sec. 31.4 we had for the complex form of the expansion

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{2\pi i k t}$$

$$c_k = \frac{a_k - ib_k}{2} \quad k > 0$$

$$c_k = \frac{a_k + ib_k}{2} \quad k < 0$$

So that

$$4|c_k|^2 = 4|c_{-k}|^2 = a_k^2 + b_k^2$$

or for real functions $f(t)$

$$|c_k| + |c_{-k}| = \sqrt{a_k^2 + b_k^2}$$

For many purposes it is the power spectrum we care about; but let us be clear: there is a loss of information in going to it. For each pair of terms we get

$$a_k \cos kt + b_k \sin kt$$

$$= \sqrt{a_k^2 + b_k^2} \left(\frac{a_k}{\sqrt{a_k^2 + b_k^2}} \cos kt + \frac{b_k}{\sqrt{a_k^2 + b_k^2}} \sin kt \right)$$

$$= \sqrt{a_k^2 + b_k^2} \cos(kt - \phi)$$

where we have picked ϕ so that

$$\frac{a_k}{\sqrt{a_k^2 + b_k^2}} = \cos \phi \quad \frac{b_k}{\sqrt{a_k^2 + b_k^2}} = \sin \phi$$

Thus the power spectrum does not include information about the phase angle ϕ .

31.8 INTERPOLATION OF PERIODIC FUNCTIONS

Part II developed the theory of polynomial approximation, and we now face the problem of developing a similar theory for the Fourier-series approximation of a continuous periodic function.

While the problem may be discussed for unequally spaced sample points, and some results are known, both theory and practice are usually based on equally

spaced points. For $2N$ equally spaced sample points the interpolation problem is solved by using the finite Fourier series to estimate the function between the sample points.

As in Part II, once we have the formula, we next want to know the error of the approximation. The error can be divided into two parts: that due to taking only a finite number of terms in the series and that due to the sampling effect (aliasing). We have already studied the effects of sampling, and we now examine the effect of taking only a finite number of terms of a Fourier series.

If the continuous Fourier series is of the usual form, then substituting the formulas for the coefficients, we get for the first $2N$ terms, where we take one-half the last cosine term as we would for a discrete series, and one-half the last sine term, since it is not identically zero in the continuous case,

$$\begin{aligned} y_{2N}(t) &= \frac{1}{N} \int_0^{2N} y(s) \left\{ \frac{1}{2} + \sum_{k=1}^{N-1} \left[\cos \frac{\pi}{N} kt \cos \frac{\pi}{N} ks + \sin \frac{\pi}{N} kt \sin \frac{\pi}{N} ks \right] \right. \\ &\quad \left. + \frac{1}{2} \sin \pi t \sin \pi s + \frac{1}{2} \cos \pi t \cos \pi s \right\} ds \\ &= \frac{1}{N} \int_0^{2N} y(s) \left\{ \frac{1}{2} + \sum_{k=1}^{N-1} \left[\cos \frac{\pi}{N} k(t-s) \right] + \frac{1}{2} \cos \pi(t-s) \right\} ds \end{aligned}$$

In order to sum¹ the quantity in the braces, we multiply by

$$\sin \frac{\pi}{2N}(t-s)$$

and get

$$\frac{1}{2} \left\{ \sin \frac{\pi}{2N} (t-s) + \left[\sin \frac{\pi}{N} \left(1 + \frac{1}{2} \right) (t-s) - \sin \frac{\pi}{N} \left(1 - \frac{1}{2} \right) (t-s) \right] + \left[\sin \frac{\pi}{N} \left(2 + \frac{1}{2} \right) (t-s) - \sin \frac{\pi}{N} \left(2 - \frac{1}{2} \right) (t-s) \right] \right. \\ \dots \dots \dots + \left[\sin \frac{\pi}{N} \left(N - 1 + \frac{1}{2} \right) (t-s) - \sin \frac{\pi}{N} \left(N - 1 - \frac{1}{2} \right) (t-s) \right] \\ \left. + \sin \frac{\pi}{2N} (t-s) \cos \pi(t-s) \right\}$$

¹ We could clearly use the methods of Chap. 11 to sum this series more elegantly, but it is likely that the reader has forgotten such details by now, and so we shall use the more clumsy but familiar trigonometric methods.

Because of cancellation of each term on the left by the term on the right one line lower, all that we have left is two terms

$$\frac{1}{2} \left\{ \sin \left[\frac{\pi}{N} \left(N - \frac{1}{2} \right) (t - s) \right] + \sin \frac{\pi}{2N} (t - s) \cos \pi(t - s) \right\}$$

We now expand the first of these terms in the form

$$\sin \pi(t - s) \cos \frac{\pi}{2N} (t - s) - \cos \pi(t - s) \sin \frac{\pi}{2N} (t - s)$$

and because of cancellation of the second of these terms with the second term that we had left, for the braces in the integral we have finally

$$\frac{\sin \pi(t - s) \cos(\pi/2N)(t - s)}{2 \sin(\pi/2N)(t - s)}$$

Thus

$$y_{2N}(t) = \frac{1}{N} \int_0^{2N} y(s) \frac{\sin \pi(t - s) \cos(\pi/2N)(t - s)}{2 \sin(\pi/2N)(t - s)} ds$$

Since $y(s)$ is assumed to be periodic, we can shift the limits to any interval of length $2N$ and set $t - s = \theta$:

$$y_{2N}(t) = \frac{1}{2N} \int_{-N}^N y(t - \theta) \frac{\sin \pi \theta \cos[(\pi/2N)\theta]}{\sin[N(\pi/2)\theta]} d\theta$$

This form is deceptive; the more samples $2N$ we take, the longer the interval. In practice, of course, the range of periodicity is fixed, and we change the sampling interval Δt . Thus we set

$$t = 2Ns$$

$$\theta = 2N\phi$$

to get the form

$$y_{2N}(s) = \int_{1/2}^{1/2} y(s - \phi) \sin(2\pi N\phi) \frac{\cos \pi\phi}{\sin \pi\phi} d\phi$$

where the change in the meaning of $y_{2N}(\cdot)$ should not cause serious confusion.

If $y(t) \equiv 1$, then we have

$$1 = \int_{1/2}^{1/2} \sin(2\pi N\phi) \frac{\cos \pi\phi}{\sin \pi\phi} d\phi$$

Since the variable of integration is ϕ , we can write

$$y(s) = \int_{1/2}^{1/2} y(s) \sin(2\pi N\phi) \frac{\cos \pi\phi}{\sin \pi\phi} d\phi$$

and subtract from this $y_{2N}(s)$ to get

$$\varepsilon_{2N}(s) = y(s) - y_{2N}(s) = \int_{-1/2}^{1/2} [y(s) - y(s - \phi)] \left\{ \sin(2N\pi\phi) \frac{\cos \pi\phi}{\sin \pi\phi} \right\} d\phi$$

The quantity inside the braces is often called¹ the *kernel*, $K(\phi)$. Clearly $K(0) = 2N$. For large N , $\sin(2N\pi\phi)$ oscillates rapidly while the term $\cot \pi\phi$ dampens (modulates) the magnitude of the oscillation until at $\phi = \pm 1/2$, $K(\pm 1/2) = 0$. The envelope of the modulation is $\cot \pi\phi$, and

$$K(\phi) = K(-\phi)$$

If the function $y(x)$ is smooth, then the major contribution to the integral, as far as the kernel is concerned, comes from the immediate neighborhood of $\phi = 0$, say from $-1/2N$ to $1/2N$, between the first zeros on each side. Thus the error in the approximation tends to be a local property (see Fig. 31.8.1). As N increases, the peak becomes higher and narrower, and the side lobe (local peaks) oscillations tend to cancel out.

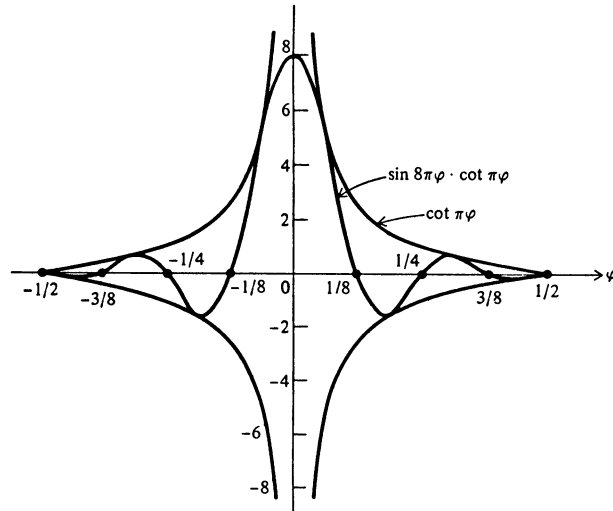


FIGURE 31.8.1

¹ See Ref. 64, p.50, where it is called a "modified kernel."

If, on the other hand, there are rapid changes in $y(x)$, then the tendency of the damped oscillations to approximately cancel out and thus contribute only small amounts is no longer present, and sudden changes in $y(x)$ remote from the point of interpolation can produce errors.

Note that the integrand involves the function rather than a high-order derivative as in polynomial approximation.

PROBLEMS

- 31.8.1 We have approximated trigonometric functions by polynomials. Approximate $y(t) = (1 - t^2)^2$ ($-1 \leq t \leq 1$) by a Fourier series using $N = 3$, that is, samples at $t = -2/3, -1/3, 0, 1/3, 2/3, 1$, and interpolate for the value at $t = 1/2$.
- 31.8.2 In this section we have developed the equivalent of Lagrange interpolation; examine the equivalent of Hermite interpolation which matches the function and derivative at each sample point. *Ans.* Not possible

31.9 INTEGRATION

As in Part II, having examined the question of interpolation and its errors, we next turn to the problem of integration. If the coefficients of the continuous Fourier series of $y(t)$ fall off fairly rapidly as $n \rightarrow \infty$, then the A_0 coefficient of the finite Fourier series gives a good estimate of the integral

$$\int_0^{2N} y(t) dt = Na_0$$

since

$$A_0 = \frac{1}{N} \sum_{i=0}^{2N-1} y(t) = a_0 + 2 \sum_{j=1}^{\infty} a_{2Nj}$$

Thus the error involved in estimating the integral from $2N$ equally spaced, equally weighted samples is

$$2N \sum_{j=1}^{\infty} a_{2Nj}$$

and represents the inevitable aliasing due to the discrete sampling.

For periodic functions the formula for integration which uses $2N$ equally spaced, equally weighted samples corresponds to the following:

Newton-Cotes, because it is equally spaced

Chebyshev, because it is equally weighted

Gauss, because $2N$ samples give exactness for $4N$ functions

Let us illustrate the accuracy of this formula for integrating periodic functions by a pair of examples. Consider the two elliptic integrals

$$I_1 = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \theta} d\theta$$

$$I_2 = \int_0^{\pi/2} \frac{1}{\sqrt{1 - k^2 \sin^2 \theta}} d\theta$$

where $|k| \leq 1$ is a parameter. We can imagine expanding the square root as a binomial. If k^2 is small, then the series will converge rapidly. Converting the powers $\sin^{2n} \theta$ to multiple angles

$$\begin{aligned} \left(\frac{e^{i\theta} - e^{-i\theta}}{2i} \right)^{2n} &= \frac{1}{(2^{2n})(i^{-2n})} [e^{2ni\theta} - C(2n, 1)e^{2(n-1)i\theta} + \cdots] \\ &= \frac{1}{(2^{2n-1})(i^{-2n})} [\cos 2n\theta - C(2n, 1) \cos(2n-2)\theta + \cdots] \end{aligned}$$

we get a rapidly converging Fourier series. Thus for small k^2 we expect accurate answers; for k^2 near 1 we do not expect accurate answers.

As an experiment, we chose sample points at 0, 30, 60, 90°, although to get periodicity we had to assume six points in the interval of periodicity, $0 \leq t \leq \pi$. Thus we actually computed only four sample points, one of which ($\theta = 0$) was trivial. The result (compared with those from a table) are given in Table 31.9.

Table 31.9 ELLIPTIC INTEGRAL $I_1(k^2)$

k^2	$I_1(\text{calc.})$	$I_1(\text{table})$	Error
$k^2 = \frac{1}{4}$	1.46746	1.4675	
$k^2 = \frac{1}{2}$	1.35064	1.3506	
$k^2 = \frac{3}{4}$	1.21099	1.2111	0.0001

ELLIPTIC INTEGRAL $I_2(k^2)$

k^2	$I_2(\text{calc.})$	$I_2(\text{table})$	Error
$k^2 = \frac{1}{4}$	1.68575	1.6858	
$k^2 = \frac{1}{2}$	1.85410	1.8541	
$k^2 = \frac{3}{4}$	2.15789	2.1565	0.0014

This example shows two things. First, it is often possible to make reasonable estimates of the size of the coefficients of a Fourier series; hence we can estimate the error of the computation *before* starting it. Second, the method of integration which uses $2N$ equally spaced, equally weighted samples gives very accurate results for periodic functions. This approach has been used to compute the Bessel function $J_0(x)$ from the formula

$$J_0(x) = \frac{1}{\pi} \int_0^\pi \cos(x \sin \phi) d\phi$$

PROBLEMS

31.9.1 Evaluate $J_0(1/2)$ from the integral form, using six samples.

31.10 THE GENERAL-OPERATOR APPROACH

In Sec. 15.4 of Part II, we described a general method (third method) for finding formulas which were exact for $1, x, \dots, x^n$. In the same way we can find formulas which are exact for

$$1, \cos \frac{\pi}{N} t, \cos \frac{2\pi}{N} t, \cos \frac{3\pi}{N} t, \dots, \cos \frac{(N-1)\pi}{N} t, \cos \pi t \\ \sin \frac{\pi}{N} t, \sin \frac{2\pi}{N} t, \sin \frac{3\pi}{N} t, \dots, \sin \frac{(N-1)\pi}{N} t$$

Such conditions are the natural ones to apply to a function that is periodic and for which we use $2N$ equally spaced samples.

Let the result of the linear operator L operating on the function $f(t)$ be $L(f)$, and let it be expressed in terms of the set of $2N$ equally spaced samples of the function $t = 0, 1, 2, \dots, 2N-1$; that is,

$$L[f(t)] = w_0 f(0) + w_1 f(1) + \dots + w_{2N-1} f(2N-1)$$

For $1, \cos \frac{\pi}{N} m, \cos \frac{2\pi}{N} m, \dots, \cos \frac{\pi(N-1)}{N} m, \cos \pi m, \sin \frac{\pi}{N} m, \dots, \sin \frac{\pi(N-1)}{N} m$, the defining equations are:

$$\begin{aligned} M_0 &= w_0 + w_1 & + w_2 & + \dots + w_{2N-1} \\ M_1 &= w_0 + w_1 \cos \frac{\pi}{N} & + w_2 \cos \frac{2\pi}{N} & + \dots + w_{2N-1} \cos \frac{2N-1}{N} \\ M_2 &= w_0 + w_1 \cos \frac{2\pi}{N} & + w_2 \cos \frac{4\pi}{N} & + \dots + w_{2N-1} \cos \frac{2N-1}{N} 2\pi \\ &\dots\dots\dots \end{aligned}$$

$$M_{N-1} = w_0 + w_1 \cos \frac{N-1}{N} \pi + w_2 \cos \frac{(N-1)2\pi}{N} + \cdots + w_{2N-1} \cos \frac{(2N-1)(N-1)\pi}{N}$$

$$M_N = w_0 - w_1 + w_2 - \cdots - w_{2N-1}$$

$$M_{N+1} = 0 + w_1 \sin \frac{\pi}{N} + w_2 \sin \frac{2\pi}{N} + \cdots + w_{2N-1} \sin \frac{(2N-1)\pi}{N}$$

.....

$$M_{2N-1} = 0 + w_1 \sin \frac{(N-1)\pi}{N} + w_2 \sin \frac{2(N-1)\pi}{N} + \cdots + w_{2N-1} \sin \frac{(2N-1)(N-1)\pi}{N}$$

Thus the matrix of the unknown weights w_i is

$$\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \cos \frac{\pi}{N} & \cos \frac{2\pi}{N} & \cdots & \cos \frac{2N-1}{N} \pi \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & \cos \frac{N-1}{N} \pi & \cos \frac{(N-1)2\pi}{N} & \cdots & \cos \frac{(N-1)(2N-1)\pi}{N} \\ 1 & -1 & 1 & \cdots & -1 \\ 0 & \sin \frac{\pi}{N} & \sin \frac{2\pi}{N} & \cdots & \sin \frac{2N-1}{N} \pi \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & \sin \frac{N-1}{N} \pi & \sin \frac{N-1}{N} 2\pi & \cdots & \sin \frac{(N-1)(2N-1)\pi}{N} \end{bmatrix}$$

Because of the orthogonality relations

$$\begin{aligned} \sum_{k=0}^{2N-1} \cos \frac{m\pi}{N} k \cos \frac{n\pi}{N} k &= N\delta_{m,n} \quad m, n \neq 0, N \\ \sum_{k=0}^{2N-1} \sin \frac{m\pi}{N} k \cos \frac{n\pi}{N} k &= 0 \\ \sum_{k=0}^{2N-1} \sin \frac{m\pi}{N} k \sin \frac{n\pi}{N} k &= N\delta_{m,n} \\ \sum_{k=0}^{2N-1} 1 &= \sum_{k=0}^{2N-1} \cos^2 \pi k = 2N \end{aligned}$$

the inverse matrix is the transpose of the matrix except that two of the columns, the first and the $(N-1)$ st, corresponding to the functions 1 and $\cos \pi m$, have been divided by 2.

$$\left(\frac{1}{N}\right) \begin{pmatrix} \frac{1}{2} & 1 & \cdots & 1 & \frac{1}{2} & 0 & \cdots & 0 \\ \frac{1}{2} & \cos \frac{\pi}{N} & \cdots & \cos \frac{N-1}{N} \pi & -\frac{1}{2} & \sin \frac{\pi}{N} & \cdots & \sin \frac{N-1}{N} \pi \\ \frac{1}{2} & \cos \frac{2\pi}{N} & \cdots & \cos \frac{N-1}{N} 2\pi & \frac{1}{2} & \sin \frac{2\pi}{N} & \cdots & \sin \frac{N-1}{N} 2\pi \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \frac{1}{2} \cos \frac{2N-1}{N} \pi & \cdots & \cos \frac{(N-1)(2N-1)}{N} \pi & -\frac{1}{2} \sin \frac{2N-1}{N} \pi & \cdots & \sin \frac{(N-1)(2N-1)}{N} \pi \end{pmatrix}$$

Here we have produced the postmultiplier inverse, while in Sec. 15.5 we produced the premultiplier inverse; they are, of course, the same, but in each case it is harder to think about the other inverse. Interchanging the order of the two matrices gives a different set of orthogonal relations, which, at times, can be useful.

This general approach of using the third method may be illustrated by an example. Suppose that we wish to estimate the first derivative of the periodic function at the point $t = 0$. To be specific, suppose that we examine the case $N = 3$. The components of the "moments" M are, using $1, \cos(\pi/3)t, \cos(2\pi/3)t, \cos \pi t, \sin(\pi/3)t, \sin(2\pi/3)t$,

$$0, 0, 0, 0, \frac{\pi}{3}, \frac{2\pi}{3}$$

The inverse matrix times the moment vector gives

$$\left(\frac{1}{3}\right) \begin{pmatrix} 1 & 2 & 2 & 1 & 0 & 0 \\ 1 & 1 & -1 & -1 & \sqrt{3} & \sqrt{3} \\ 1 & -1 & -1 & 1 & \sqrt{3} & -\sqrt{3} \\ 1 & -2 & 2 & -1 & 0 & 0 \\ 1 & -1 & -1 & 1 & -\sqrt{3} & \sqrt{3} \\ 1 & 1 & -1 & -1 & -\sqrt{3} & -\sqrt{3} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \frac{\pi}{3} \\ \frac{2\pi}{3} \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 0 \\ \sqrt{3} \pi \\ -\frac{\sqrt{3}}{3} \pi \\ 0 \\ \frac{\sqrt{3}}{3} \pi \\ -\sqrt{3} \pi \end{pmatrix}$$

The last vector is the weight vector; thus the formula is

$$f'(0) = \frac{\sqrt{3} \pi}{18} [3f(1) - f(2) + f(4) - 3f(5)]$$

PROBLEMS

31.10.1 Using the matrix of the illustrative example, find a formula for $f''(0)$.

31.10.2 Using the matrix, find a formula for estimating $\int_0^{2N} \int_0^t f(t) dt dt$.

31.11 SOME REMARKS ON THE GENERAL METHOD

In the case of polynomial approximation, it was efficient to use the general matrix approach. In the case of approximation with a Fourier series, it is not. The difference is due to the fact that we can find the Fourier expansion of a function with much less labor than performing the matrix multiplication to get the weights (see Sec. 31.5). Once we have the Fourier coefficients of the function, it is easy to multiply them by the moments to get the final answer. Thus if we have the Fourier series

$$f(t) = \frac{a_0}{2} + \sum_{k=1}^{N-1} a_k \cos \frac{\pi}{N} kt + \frac{a_N}{2} \cos \pi t + \sum_{k=1}^{N-1} b_k \sin \frac{\pi}{N} kt$$

applying the operator $L(\cdot)$, we have

$$\begin{aligned} L[f(t)] &= \frac{a_0}{2} L(1) + \sum_{k=1}^{N-1} a_k L\left(\cos \frac{\pi}{N} kt\right) + \frac{a_N}{2} L(\cos \pi t) \\ &\quad + \sum_{k=1}^{N-1} b_k L\left(\sin \frac{\pi}{N} kt\right) \\ &= \frac{a_0}{2} M_0 + \sum_{k=1}^{N-1} a_k M_k + \frac{a_N}{2} M_N + \sum_{k=1}^{N-1} b_k M_{N+k} \end{aligned}$$

The reason that we can find the coefficients easily is mainly that the sines and cosines are orthogonal and can be found independently of each other. Thus the inverse-matrix approach, while of theoretical interest for Fourier-series approximation, is not of much practical importance.

It is necessary to point out again a distinction which first arose in Sec. 21.3, which covers indefinite integration. Whether we make a formula such as

$$y = \int_a^b f(x) dx$$

exact for $f(x) = 1, x, x^2, \dots$ or for $y = 1, x, x^2, \dots$ makes a difference. As was pointed out in Sec. 21.3, exactness for $f(x) = 1, x, x^2, \dots$ is the same, for this formula, as exactness for $y = x, x^2, x^3, \dots$ since the integrals of powers of x are powers of x one higher. We found it convenient at that time to *add* the condition of exactness for $y = 1$ and thus restore the equivalence of the two choices.

For the Fourier-series approach, if we first approximate the integrand and then integrate, the terms $\cos kx$ and $\sin kx$ go into each other, but the constant term 1 goes into x . Thus the equivalence of approximating the integrand or the solution by a Fourier series is lost. We must, therefore, decide *before we start* on what part of the problem we are going to approximate by the set of functions we are using—approximations at various stages of the problem are not necessarily equivalent. In practice it is usual to know properties of the input function or the output function, and hence these are the ones usually chosen for the approximation.

This effect of nonequivalence will be seen again in Chaps. 39 and 40 on exponential functions where, upon integration, the exponential $e^{\alpha_i x}$ goes into itself (within a multiplicative constant) *except* $\alpha_i = 0$.

CONVERGENCE OF FOURIER SERIES

32.1 THE IMPORTANCE OF CONVERGENCE

The rate of convergence of a Fourier series is important for two reasons. First it enables us to *estimate in advance* the sampling rate we will need to use in the computation to avoid the errors that arise from the inevitable aliasing due to sampling the function at equally spaced points. As we shall show, this rate can be reasonably well estimated from the appearance of the function—the singularities along the real axis control the rate of the convergence, rather than, as in the polynomial case, the size of the higher-order derivatives, depending on the location of the singularities in the complex plane.

Second, we shall examine methods for improving the rate of convergence, so that either lower sampling rates can be used or a higher accuracy can be obtained using the same rate. Thus a great deal of machine time can often be saved. We shall give two approaches to this important topic, one applicable before numerically finding the Fourier coefficients A_k and B_k and the other after they are found.

32.2 STRAIGHT-LINE APPROXIMATION

Many times the function we wish to use can be approximated by a sequence of straight lines. We examine two cases: first the lines are continuous (remember that the function is periodic and that the two end values are the same point), and second the function is approximated by a broken straight line. We shall suppose that in the i th interval the function is approximated by (see Fig. 32.2.1)

$$F(x) = \alpha_i + \beta_i x \quad x_{i-1} < x < x_i \\ i = 1, 2, \dots, K$$

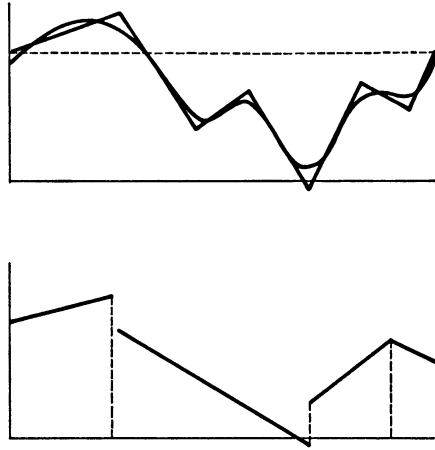


FIGURE 32.2.1

We have in the first case

$$\begin{aligned} a_k &= \frac{1}{\pi} \int_0^{2\pi} F(x) \cos kx \, dx = \frac{1}{\pi} \sum_{i=1}^K \int_{x_{i-1}}^{x_i} (\alpha_i + \beta_i x) \cos kx \, dx \\ &= \frac{1}{\pi} \sum_{i=1}^K \left[(\alpha_i + \beta_i x) \frac{\sin kx}{k} \Big|_{x_{i-1}}^{x_i} - \int_{x_{i-1}}^{x_i} \beta_i \frac{\sin kx}{k} \, dx \right] \end{aligned}$$

Due to continuity the integrated part drops out, and we get

$$a_k = -\frac{1}{\pi} \sum_{i=1}^K \beta_i \frac{\cos kx}{k^2} \Big|_{x_{i-1}}^{x_i}$$

Let M_1 be the maximum of the $|\beta_i|$. Then

$$|a_k| < \frac{M_1}{\pi} \frac{2K}{k^2} = \frac{2M_1}{k^2} \frac{K}{\pi}$$

Similarly for the b_k . Thus the series converges like $1/k^2$.

In the second case, the broken straight line, the integrated part does *not* drop out, and we have convergence like $1/k$.

Thus we see that the behavior along the real axis tells us the rate of convergence. For practical purposes the converse is true; if the coefficients fall off like $1/k$, then there are one or more simple discontinuities; if they fall off like $1/k^2$, then there are no discontinuities.¹

PROBLEMS

Find the rate of convergence of the Fourier series for

$$32.2.1 \quad F(t) = \begin{cases} 0 & 0 < t < \pi \\ 1 & \pi < t < 2\pi \end{cases}$$

$$32.2.2 \quad F(t) = t \quad -\pi < t < \pi$$

$$32.2.3 \quad F(t) = |t| \quad -\pi < t < \pi$$

$$32.2.4 \quad F(t) = \begin{cases} 0 & 0 < t < a \\ 1 & a < t < 2\pi \end{cases}$$

32.3 FUNCTIONS HAVING CONTINUOUS HIGHER DERIVATIVES

If a function has a continuous first derivative and, except for a finite number of places, a continuous second derivative, then we can proceed as in Sec. 32.2 except that we integrate by parts twice

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} F(t) \cos kt \, dt = -\frac{1}{\pi k} \int_{-\pi}^{\pi} F'(t) \sin kt \, dt$$

the continuity causing, as before, the integrated part to cancel out. Thus the assumption of straight lines is not essential to the argument of Sec. 32.2. We again break up the range into pieces in which $F''(t)$ is continuous

$$\begin{aligned} a_k &= -\frac{1}{\pi k} \sum_{i=0}^K \int_{t_{i-1}}^{t_i} F'(t) \sin kt \, dt \\ &= -\frac{1}{\pi k^2} \sum_{i=0}^K \int_{t_{i-1}}^{t_i} F''(t) \cos kt \, dt \\ |a_k| &\leq \frac{M_2}{\pi k^2} 2K = \frac{2M_2}{k^2} \frac{K}{\pi} \end{aligned}$$

where M_2 is the maximum of $|F''(t)|$. We treat b_k similarly.

¹ G. Raisbeck, Order of Magnitude of Fourier Coefficients, *Am. Math. Monthly*, vol. 62, pp. 149–154, March, 1955.